

CHAPTER 6

TSUNAMI LOADS AND EFFECTS

6.1 GENERAL REQUIREMENTS

6.1.1 Scope. The following buildings and other structures located within the Tsunami Design Zone shall be designed for the effects of Maximum Considered Tsunami, including hydrostatic and hydrodynamic forces, waterborne debris accumulation and impact loads, subsidence, and scour effects in accordance with this chapter:

- Tsunami Risk Category IV buildings and structures;
- Tsunami Risk Category III buildings and structures with inundation depth greater than 3 ft (0.914 m) at any location within the intended footprint of the structure, and
- Where required by a state or locally adopted building code statute to include design for tsunami effects, Tsunami Risk Category II buildings with mean height above grade plane greater than the height designated in the statute and having inundation depth greater than 3 ft (0.914 m) at any location within the intended footprint of the structure.

EXCEPTION: Tsunami Risk Category II single-story buildings of any height without mezzanines or any occupiable roof level and not having any critical equipment or systems need not be designed for the tsunami loads and effects specified in this chapter.

For the purposes of this chapter, Tsunami Risk Category shall be as determined in accordance with Section 6.4.

Tsunami Design Zone shall be determined using the ASCE Tsunami Design Geodatabase of geocoded reference points shown in Fig. 6.1-1. The ASCE Tsunami Design Geodatabase of geocoded reference points of Runup and associated Inundation Limits of the Tsunami Design Zone is available at [<http://asce7tsunami.online>].

EXCEPTION: For coastal regions subject to tsunami inundation and not covered by Fig. 6.1-1, Tsunami Design Zone, inundation limits, and runup elevations shall be determined using the site-specific procedures of Section 6.7, or for Tsunami Risk Category II or III structures, determined in accordance with the procedures of Section 6.5.1.1 using Fig. 6.7-1.

Designated nonstructural components and systems associated with Tsunami Risk Category III Critical Facilities and Tsunami Risk Category IV structures subject to this chapter shall be located above, protected from, or otherwise designed for inundation in accordance with Section 6.15 so that they are able to provide their essential functions immediately following the Maximum Considered Tsunami event.

User Note: The ASCE Tsunami Design Geodatabase of geocoded reference points of Runup and associated Inundation Limits of the Tsunami Design Zone is available at [<http://asce7tsunami.online>]. Sea level rise has not been incorporated into the Tsunami Design Zone maps, and any additive effect on the inundation at the site should be explicitly evaluated.

6.2 DEFINITIONS

The following definitions apply only to the tsunami requirements of this chapter. Also see Fig. 6.2-1 for an illustration of some key terms.

ASCE TSUNAMI DESIGN GEODATABASE: The ASCE database (version 2016-1.0) of geocoded reference points of Offshore 328-ft (100-m) depth Tsunami Amplitude, H_T , and Predominant Period, T_{TSU} , of the Maximum Considered Tsunami, disaggregated hazard source contribution figures, probabilistic subsidence, Runup Elevation and Inundation geocoded reference points, and Tsunami Design Zone maps.

BATHYMETRIC PROFILE: A cross section showing ocean depth plotted as a function of horizontal distance from a reference point (such as a coastline).

CHANNELIZED SCOUR: Scour that results from broad flow that is diverted to a focused area such as return flow in a preexisting stream channel or alongside a seawall.

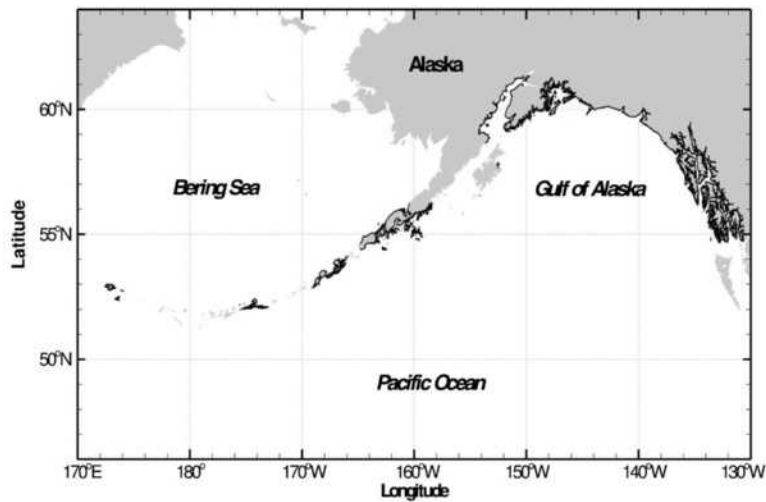
CLOSURE RATIO (OF INUNDATED PROJECTED AREA): Ratio of the area of enclosure, not including glazing and openings, that is inundated to the total projected vertical plane area of the inundated enclosure surface exposed to flow pressure.

COLLAPSE PREVENTION STRUCTURAL PERFORMANCE LEVEL: A postevent damage state in which a structure has damaged components and continues to support gravity loads but retains little or no margin against collapse.

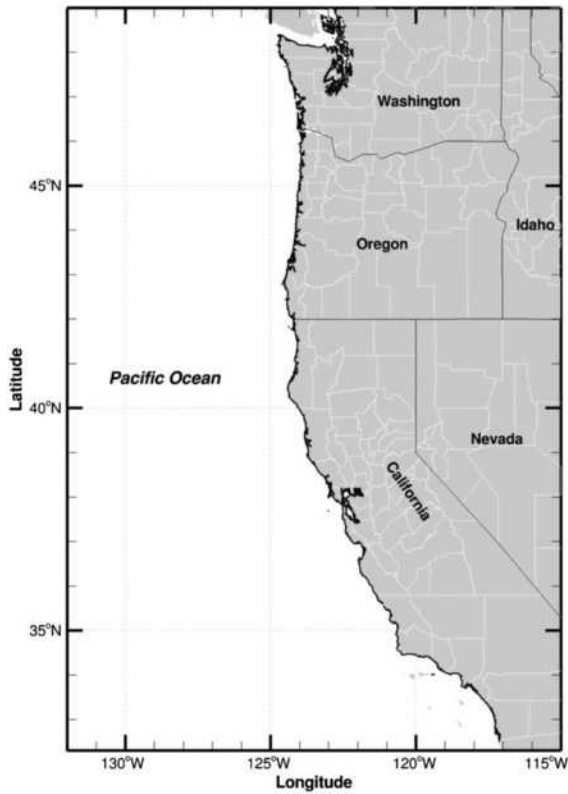
CRITICAL EQUIPMENT OR CRITICAL SYSTEMS: Nonstructural components designated essential for the functionality of the critical facility or essential facility or that are necessary to maintain safe containment of hazardous materials.

CRITICAL FACILITY: Buildings and structures that provide services that are designated by federal, state, local, or tribal governments to be essential for the implementation of the response and recovery management plan or for the continued functioning of a community, such as facilities for power, fuel, water, communications, public health, major transportation infrastructure, and essential government operations. Critical facilities comprise all public and private facilities deemed by a community to be essential for the delivery of vital services, protection of special populations, and the provision of other services of importance for that community.

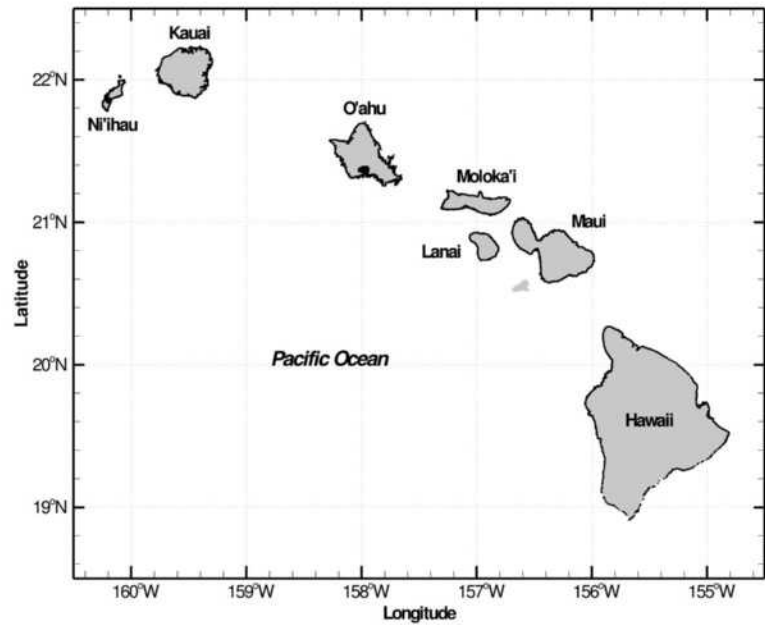
DEADWEIGHT TONNAGE (DWT): Deadweight Tonnage (DWT) is a vessel's Displacement Tonnage (DT) minus its Lightship Weight (LWT). DWT is a classification used for the carrying capacity of a vessel that is equal to the sum of the weights of cargo, fuel, fresh water, ballast water, provisions, passengers, and crew; it does not include the weight of the vessel itself. Displacement Tonnage is the total weight of a fully loaded vessel. Lightship Weight is the weight of the vessel without cargo, crew, fuel, fresh water, ballast water, provisions, passengers, or crew.



(a) Alaska



(b) Pacific Coast



(c) Hawaii

FIGURE 6.1-1 Extent of ASCE Tsunami Design Geodatabase¹ of Geocoded Reference Points of Runup and Associated Inundation Limits of the Tsunami Design Zone²

¹The ASCE Tsunami Design Geodatabase of geocoded reference points of Runup and associated Inundation Limits of the Tsunami Design Zone is available at <http://asce7tsunami.online>.

²Sea level rise has not been incorporated into the Tsunami Design Zone maps, and any additive effect on the inundation at the site should be explicitly evaluated.

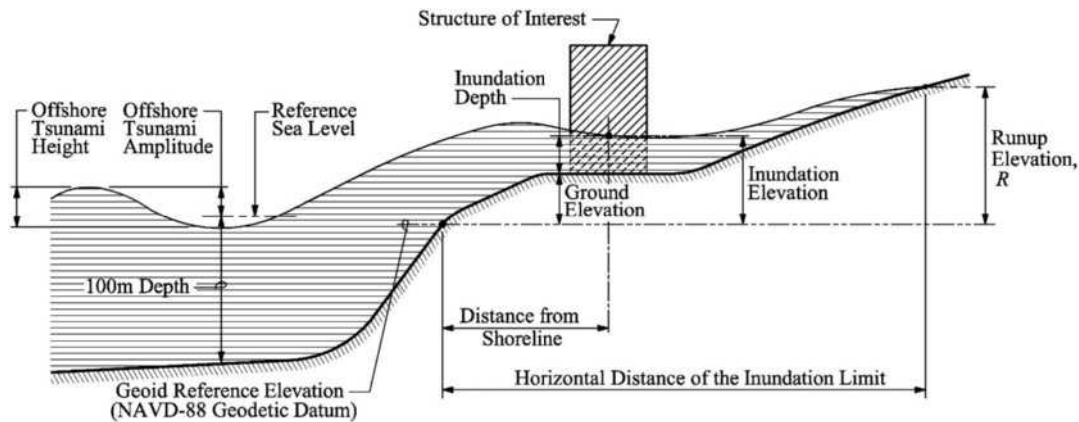


FIGURE 6.2-1 Illustration of Key Definitions along a Flow Transect in a Tsunami Design Zone

DESIGN STRENGTH: Nominal strength multiplied by a resistance factor, ϕ .

DESIGN TSUNAMI PARAMETERS: The tsunami parameters used for design, consisting of the inundation depths and flow velocities at the stages of inflow and outflow most critical to the structure and momentum flux.

DESIGNATED NONSTRUCTURAL COMPONENTS AND SYSTEMS: Nonstructural components and systems that are assigned a component Importance Factor, I_p , equal to 1.5 per Section 13.1.3 of this standard.

DUCTILITY-GOVERNED ACTION: Any action on a structural component characterized by postelastic force versus deformation curve that has (1) sufficient ductility and (2) results from an impulsive short-term force that is not sustained.

FORCE-SUSTAINED ACTIONS: Any action on a structural component characterized by a sustained force or a postelastic force versus deformation curve that is not ductility-governed due to lack of sufficient ductility.

FROUDE NUMBER, F_r : A dimensionless number defined by $u/\sqrt{gh}$, where u is the flow velocity averaged over the cross section perpendicular to the flow, which is used to quantify the normalized tsunami flow velocity as a function of water depth.

GENERAL EROSION: A general wearing away and erosion of the land surface over a significant portion of the inundation area, excluding localized scour actions.

GRADE PLANE: A horizontal reference plane at the site representing the average elevation of finished ground level adjoining the structure at all exterior walls. Where the finished ground level slopes away from the exterior walls, the grade plane is established by the lowest points within the area between the structure and the property line or, where the property line is more than 6 ft (1.83 m) from the structure, between the structure and points 6 ft (1.83 m) from the structure.

HAZARD-CONSISTENT TSUNAMI SCENARIO: One or more surrogate tsunami scenarios generated from the principal disaggregated seismic source regions that replicate the offshore tsunami waveform characteristics of Fig. 6.7-1 and Table 6.7-1 for the site of interest, taking into account the net effect of the probabilistic treatment of uncertainty into the offshore wave amplitude of the scenario(s).

HYDRODYNAMIC LOADS: Loads imposed on an object by water flowing against and around it.

HYDROSTATIC LOADS: Loads imposed on an object by a standing mass of water.

IMMEDIATE OCCUPANCY STRUCTURAL PERFORMANCE LEVEL: The postevent damage state in which a structure remains safe to occupy.

IMPACT LOADS: Loads that result from debris or other object transported by the design tsunami striking a structure or portion thereof.

INUNDATION DEPTH: The depth of design tsunami water level, including relative sea level change, with respect to the grade plane at the structure.

INUNDATION ELEVATION: The elevation of the design tsunami water surface, including relative sea level change, with respect to vertical datum in North American Vertical Datum (NAVD 88).

INUNDATION LIMIT: The maximum horizontal inland extent of flooding for the Maximum Considered Tsunami, where the inundation depth above grade becomes zero; the horizontal distance that is flooded, relative to the shoreline defined where the North American Vertical Datum of 1988 (NAVD 88) elevation is zero.

LIFE SAFETY STRUCTURAL PERFORMANCE LEVEL: The postevent damage state is that in which a structure has damaged components but retains a margin against onset of partial or total collapse.

LIQUEFACTION SCOUR: The limiting case of pore pressure softening associated with hydrodynamic flow, where the effective stress drops to zero. In noncohesive soils, the shear stress required to initiate sediment transport also drops to zero during liquefaction scour.

LOCAL COSEISMIC TSUNAMI: A tsunami preceded by an earthquake with damaging effects felt within the subsequently inundated area.

LOCAL SCOUR: Removal of material from a localized portion of land surface, resulting from flow around, over, or under a structure or structural element.

MAXIMUM CONSIDERED TSUNAMI: A probabilistic tsunami having a 2% probability of being exceeded in a 50-year period or a 2,475-year mean recurrence interval.

MOMENTUM FLUX: The quantity $\rho_s h u^2$ for a unit width based on the depth-averaged flow speed u , over the inundation depth h , for equivalent fluid density ρ_s , having the units of force per unit width.

NEARSHORE PROFILE: Cross-sectional bathymetric profile from the shoreline to a water depth of 328 ft (100 m).

NEARSHORE TSUNAMI AMPLITUDE: The Maximum Considered Tsunami amplitude immediately off the coastline at 33 ft (10 m) of water depth.

NONBUILDING CRITICAL FACILITY STRUCTURE: Nonbuilding structure whose Tsunami Risk Category is designated as either III or IV.

NONBUILDING STRUCTURE: A structure other than a building.

OFFSHORE TSUNAMI AMPLITUDE: Maximum Considered Tsunami amplitude relative to the Reference Sea Level, measured where the undisturbed water depth is 328 ft (100 m).

OFFSHORE TSUNAMI HEIGHT: Waveform vertical dimension of the Maximum Considered Tsunami from consecutive trough to crest, measured where the undisturbed water depth is 328 ft (100 m), after removing the tidal variation.

OPEN STRUCTURE: A structure in which the portion within the inundation depth has no greater than 20% closure ratio, and in which the closure does not include any Tsunami Breakaway Walls, and which does not have interior partitions or contents that are prevented from passing through and exiting the structure as unimpeded waterborne debris.

PILE SCOUR: A special case of enhanced local scour that occurs at a pile, bridge pier, or similar slender structure.

PLUNGING SCOUR: A special case of enhanced local scour that occurs when the flow passes over a complete or nearly complete obstruction, such as a barrier wall, and drops steeply onto the ground below, scouring out a depression.

PORE PRESSURE SOFTENING: A mechanism that enhances scour through increased pore-water pressure generated within the ground during rapid tsunami loading and the release of that pressure during drawdown.

PRIMARY STRUCTURAL COMPONENT: Structural components required to resist tsunami forces and actions and inundated structural components of the gravity-load-carrying system.

RECOGNIZED LITERATURE: Published research findings and technical papers that are approved by the Authority Having Jurisdiction.

REFERENCE SEA LEVEL: The sea level datum used in site-specific inundation modeling that is typically taken to be Mean High Water Level (MHWL).

RELATIVE SEA LEVEL CHANGE: The local change in the level of the ocean relative to the land, which might be caused by ocean rise and/or subsidence of the land.

RUNUP ELEVATION: Ground elevation at the maximum tsunami inundation limit, including relative sea level change, with respect to the North American Vertical Datum 1988 (NAVD 88) reference datum.

SECONDARY STRUCTURAL COMPONENT: A structural component that is not primary.

SHOALING: The increase in wave height and wave steepness caused by the decrease in water depth as a wave travels into shallower water.

SOLITON FISSION: Short-period waves generated on the front edge of a tsunami waveform under conditions of shoaling on a long and gentle seabed slope or having abrupt seabed discontinuities, such as fringing reefs.

STRUCTURAL COMPONENT: A component of a building that provides gravity-load-carrying or lateral-force resistance as part of a continuous load path to foundation, including beams, columns, slabs, braces, walls, wall piers, coupling beams, and connections.

STRUCTURAL WALL: A wall that provides gravity-load-carrying support or one that is designed to provide lateral-force resistance.

SURGE: Rapidly rising water level resulting in horizontal flow inland.

SUSTAINED FLOW SCOUR: Enhanced local scour that results from flow acceleration around a structure. The flow acceleration and associated vortices increase the bottom shear stress and scour out a localized depression.

TOE SCOUR: A special case of enhanced local scour that occurs at the base of a seawall or similar structure on the side

directly exposed to the flow. Toe scour can occur whether or not the structure is overtopped.

TOPOGRAPHIC TRANSECT: Profile of vertical elevation data versus horizontal distance along a cross section of the terrain, in which the orientation of the cross section is perpendicular or at some specified orientation angle to the shoreline.

TSUNAMI: A series of waves with variable long periods, typically resulting from earthquake-induced uplift or subsidence of the seafloor.

TSUNAMI AMPLITUDE: The absolute value of the difference between a particular peak or trough of the tsunami and the undisturbed sea level at the time.

TSUNAMI BORE: A steep and turbulent broken wavefront generated on the front edge of a long-period tsunami waveform when shoaling over mild seabed slopes or abrupt seabed discontinuities such as fringing reefs, or in a river estuary, per Section 6.6.4. Soliton fission in the Nearshore Profile can often lead to the occurrence of tsunami bores.

TSUNAMI BORE HEIGHT: The height of a broken tsunami surge above the water level in front of the bore or grade elevation if the bore arrives on nominally dry land.

TSUNAMI BREAKAWAY WALL: Any type of wall subject to flooding that is not required to provide structural support to a building or other structure and that is designed and constructed such that, before the development of the design flow conditions of Inundation Load Case 1, as defined in Section 6.8.3.1, the wall will collapse or detach in such a way that (1) it allows substantially free passage of floodwaters and external or internal waterborne debris, including unattached building contents and (2) it does not damage the structure or supporting foundation system.

TSUNAMI DESIGN ZONE: An area identified on the Tsunami Design Zone Map between the shoreline and the inundation limit, within which structures are analyzed and designed for inundation by the Maximum Considered Tsunami.

TSUNAMI DESIGN ZONE MAP: The map given in Fig. 6.1-1 designating the potential horizontal inundation limit of the Maximum Considered Tsunami, or a state or local jurisdiction's probabilistic map produced in accordance with Section 6.7 of this chapter.

TSUNAMI EVACUATION MAP: An evacuation map based on a tsunami inundation map based on assumed scenarios that is developed and provided to a community by either the applicable state agency or NOAA under the National Tsunami Hazard Mitigation Program. Tsunami inundation maps for evacuation may be significantly different in extent than the Probabilistic Tsunami Design Zone, and Tsunami Evacuation Maps are not intended for design or land use purposes.

TSUNAMI-PRONE REGION: The coastal region in the United States addressed by this chapter with quantified probability in the recognized literature of tsunami inundation hazard with runup greater than 3 ft (0.914 m) caused by tsunamigenic earthquakes in accordance with the Probabilistic Tsunami Hazard Analysis method given in this chapter.

TSUNAMI RISK CATEGORY: The Risk Category from Section 1.5, as modified for specific use related to this chapter per Section 6.4.

TSUNAMI VERTICAL EVACUATION REFUGE STRUCTURE: A structure designated and designed to serve as a point of refuge to which a portion of the community's population can evacuate above a tsunami when high ground is not available.

6.3 SYMBOLS AND NOTATION

A_{beam} = vertical projected area of an individual beam element	I_{tsu} = importance Factor for Tsunami forces to account for additional uncertainty in estimated parameters
A_{col} = vertical projected area of an individual column element.	k = effective stiffness of the impacting debris or the lateral stiffness of the impacted structural element
A_d = vertical projected area of obstructing debris accumulated on structure	k_s = fluid density factor to account for suspended soil and other smaller flow-embedded objects that are not considered in Section 6.11
A_{wall} = vertical projected area of an individual wall element	L = live load
a_1 = amplitude of the leading pulse (negative for a leading depression Tsunami)	L_{refuge} = public assembly live load effect in the Tsunami refuge floor area
a_2 = amplitude of the following, or second, pulse	l_w = length of a structural wall
b = width subject to force	LWT = lightship Weight of vessel
B = overall building width	m = component demand modification factor accounting for expected ductility, applied to the expected strength of a ductility-governed element action, to obtain the acceptable structural component capacity at a particular performance level when using a linear static analysis procedure
C_{bs} = force coefficient with breakaway slab	m_{contents} = mass of contents in a shipping container
C_{cx} = proportion of closure coefficient	MCT = maximum Considered Tsunami
C_d = drag coefficient based on quasi-steady forces	m_d = mass of debris object
C_{dis} = discharge coefficient for overtopping	n = Manning's coefficient
C_o = orientation coefficient (of debris)	P_u = uplift pressure on slab or building horizontal element
c_{2V} = plunging scour coefficient	P_{ur} = reduced uplift pressure for slab with opening
D = dead load	q = discharge per unit width over an overtopped structure
D_a = diameter of rock armor	Q_{CE} = expected strength of the structural element
d_d = additional drop in grade to the base of wall on the side of a seawall or freestanding retaining wall subject to plunging scour	Q_{CS} = specified strength of the structural element
D_s = scour depth	Q_{UD} = ductility-governed force caused by gravity and Tsunami loading
DT = displacement Tonnage	Q_{UF} = maximum force generated in the element caused by gravity and Tsunami loading
DWT = deadweight Tonnage of vessel	R = mapped Tsunami runup elevation
E = earthquake load	R_{max} = dynamic response ratio
E_g = hydraulic head in the Energy Grade Line Analysis	R_s = net upward resistance from foundation elements
E_{mh} = horizontal Seismic Load Effect, including over-strength factor, defined in Section 12.4.3.1	s = friction slope of the energy grade line
F_d = drag force on an element or component	S = snow load
F_{dx} = drag force on the building or structure at each level	t = time
F_h = unbalanced hydrostatic lateral force	t_d = duration of debris impact
F_i = debris impact design force	TDZ = Tsunami Design Zone
F_{ni} = nominal maximum instantaneous debris impact force	t_o = offset time of the wave train
F_{pw} = hydrodynamic force on a perforated wall	T_{Tsu} = predominant wave period, or the time from the start of the first pulse to the end of the second pulse
F_r = froude number = $u/\sqrt{gh}$	u = Tsunami flow velocity
F_{Tsu} = Tsunami load or effect	U = jet velocity of plunging flow
f_{uw} = equivalent uniform lateral force per unit width	u_{max} = maximum Tsunami flow velocity at the structure
F_v = buoyancy force	u_v = vertical component of Tsunami flow velocity
F_w = load on wall or pier	V_w = displaced water volume
$F_{w\theta}$ = force on a wall oriented at an angle θ to the flow	w_g = width of opening gap in slab
g = acceleration caused by gravity	W_s = weight of the structure
h = Tsunami inundation depth above grade plane at the structure	x = horizontal distance inland from NAVD 88 shoreline
H_B = barrier height of a levee, seawall, or freestanding retaining wall	x_R = mapped inundation limit distance inland from NAVD 88 shoreline
h_e = inundated height of an individual element	z = ground elevation above NAVD 88 datum
h_i = inundation depth at point i	α = froude number coefficient in the Energy Grade Line Analysis
h_{max} = maximum inundation depth above grade plane at the structure	β = effective wake angle downstream of an obstructing structure to the structure of interest
h_o = offshore water depth	γ_s = minimum fluid weight density for design hydrostatic loads
H_O = depth to which a barrier is overtopped above the barrier height	γ_{sw} = effective weight density of seawater
h_r = residual water height within a building	Δx_i = incremental distance used in the Energy Grade Line Analysis
h_s = height of structural floor slab above grade plane at the structure	ξ_{100} = surf similarity parameter using 328 ft (100 m) near-shore wave characteristics
h_{ss} = height of the bottom of the structural floor slab, taken above grade plane at the structure	
h_{sx} = story height of story x	
H_T = offshore Tsunami Amplitude determined from Fig. 6.7-1	
H_{Tsu} = load caused by Tsunami-induced lateral earth pressure under submerged conditions	

η = free surface elevation as a function of time, t , used to drive the offshore boundary condition at the 328-ft (100-m) depth contour
 θ = angle between the longitudinal axis of a wall and the flow direction
 ϕ = structural resistance factor
 ρ_s = minimum fluid mass density for design hydrodynamic loads
 ρ_{sw} = effective mass density of seawater
 φ = average slope of grade at the structure
 φ_i = average slope of grade at point i
 Φ = mean slope angle of the Nearshore Profile
 ψ = angle between the plunging jet at the scour hole and the horizontal
 Ω = angular frequency of the waveform, equal to $2\pi/T$, where T is the wave period
 Ω_o = overstrength factor for the lateral-force-resisting system given in Table 12.2-1.

6.4 TSUNAMI RISK CATEGORIES

For the purposes of this chapter, Tsunami Risk Categories for buildings and other structures shall be the Risk Categories given in Section 1.5 with the following modifications:

1. Federal, state, local, or tribal governments shall be permitted to include Critical Facilities in Tsunami Risk Category III, such as power-generating stations, water-treatment facilities for potable water, wastewater-treatment facilities, and other public utility facilities not included in Risk Category IV.
2. The following structures need not be included in Tsunami Risk Category IV, and state, local, or tribal governments shall be permitted to designate them as Tsunami Risk Category II or III:
 - a. Fire stations, ambulance facilities, and emergency vehicle garages;
 - b. Earthquake or hurricane shelters;
 - c. Emergency aircraft hangars; and
 - d. Police stations that do not have holding cells and that are not uniquely required for postdisaster emergency response as a Critical Facility.
3. Tsunami Vertical Evacuation Refuge Structures shall be included in Tsunami Risk Category IV.

6.5 ANALYSIS OF DESIGN INUNDATION DEPTH AND FLOW VELOCITY

6.5.1 Tsunami Risk Category II and III Buildings and Other Structures. The Maximum Considered Tsunami inundation depth and tsunami flow velocity characteristics at a Tsunami Risk Category II or III building or other structure shall be determined by using the Energy Grade Line Analysis of Section 6.6 using the inundation limit and runup elevation of the Maximum Considered Tsunami given in Fig. 6.1-1.

The site-specific Probabilistic Tsunami Hazard Analysis (PTHA) in Section 6.7 shall be permitted as an alternate to the Energy Grade Line Analysis. Site-specific velocities determined by PTHA shall be subject to the limitation in Section 6.7.6.8.

EXCEPTION: For Tsunami-Prone Regions not covered by Fig. 6.1-1, the procedures of Section 6.5.1.1 shall apply to Tsunami Risk Category II and III buildings and other structures.

6.5.1.1. Runup Evaluation for Areas Where No Map Values Are Given. For Tsunami Risk Category II and III buildings and other structures where no mapped inundation limit is shown

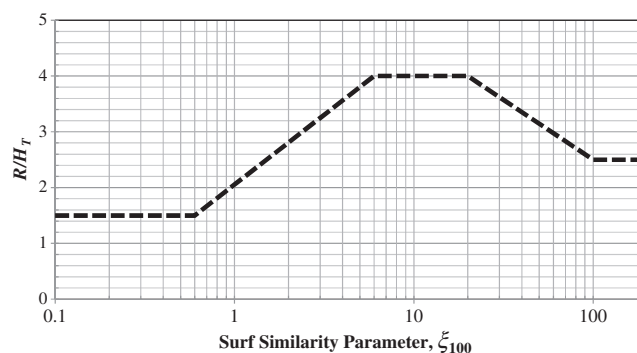


FIGURE 6.5-1. Runup Ratio R/H_T , as a Function of the Mean Slope of the Surf Similarity Parameter ξ_{100} , Where No Mapped Inundation Limit Exists

in Fig. 6.1-1, the ratio of tsunami runup elevation above Mean High Water Level to Offshore Tsunami Amplitude, R/H_T , shall be permitted to be determined using the surf similarity parameter ξ_{100} , according to Eqs. (6.5-2a, b, c, d, or e) and Fig. 6.5-1.

Surf similarity parameter, ξ_{100} , for this application to tsunami engineering shall be determined in accordance with Eq. (6.5-1).

$$\xi_{100} = \frac{T_{TSU}}{\cot \Phi} \sqrt{\frac{g}{2\pi H_T}} \quad (6.5-1)$$

where Φ is the mean slope angle of the Nearshore Profile taken from the 328-ft (100-m) water depth to the Mean High Water elevation along the axis of the topographic transect for the site. H_T is the Offshore Tsunami Amplitude, and T_{TSU} is the wave period of the tsunami at 328-ft (100-m) water depth. H_T and T_{TSU} are given in Fig. 6.7-1.

$$\text{For } \xi_{100} \leq 0.6, \quad R/H_T = 1.5 \quad (6.5-2a)$$

$$\begin{aligned} &\text{For } \xi_{100} > 0.6 \text{ and } \xi_{100} \leq 6, \\ &R/H_T = 2.50[\log_{10}(\xi_{100})] + 2.05 \end{aligned} \quad (6.5-2b)$$

$$\text{For } \xi_{100} > 6 \text{ and } \xi_{100} \leq 20, \quad R/H_T = 4.0 \quad (6.5-2c)$$

$$\begin{aligned} &\text{For } \xi_{100} > 20 \text{ and } \xi_{100} \leq 100, \\ &R/H_T = -2.15[\log_{10}(\xi_{100})] + 6.80 \end{aligned} \quad (6.5-2d)$$

$$\text{For } \xi_{100} > 100, \quad R/H_T = 2.5 \quad (6.5-2e)$$

EXCEPTION: These equations shall not be used where there is an expectation of wave focusing such as at headlands, in V-shaped bays, or where the on-land flow fields are expected to vary significantly in the direction parallel to the shoreline because of longshore variability of topography.

6.5.2 Tsunami Risk Category IV Buildings and Other Structures. The Energy Grade Line Analysis of Section 6.6 shall be performed for Tsunami Risk Category IV buildings and other structures, and the site-specific Probabilistic Tsunami Hazard Analysis (PTHA) of Section 6.7 shall also be performed. Site-specific velocities determined by site-specific PTHA determined to be less than the Energy Grade Line Analysis shall be subject to the limitation in Section 6.7.6.8.

Site-specific velocities determined to be greater than the Energy Grade Line Analysis shall be used.

EXCEPTION: For structures other than Tsunami Vertical Evacuation Refuge Structures, a site-specific Probabilistic Tsunami Hazard Analysis need not be performed where the inundation depth resulting from the Energy Grade Line Analysis is determined to be less than 12 ft (3.66 m) at any point within the location of the Tsunami Risk Category IV structure.

6.5.3 Sea Level Change. The direct physical effects of potential relative sea level change shall be considered in determining the maximum inundation depth during the project lifecycle. A project lifecycle of not less than 50 years shall be used. The minimum rate of potential relative sea level change shall be the historically recorded sea level change rate for the site. The potential increase in relative sea level during the project lifecycle of the structure shall be added to the Reference Sea Level and to the tsunami runup elevation.

6.6 INUNDATION DEPTHS AND FLOW VELOCITIES BASED ON RUNUP

6.6.1 Maximum Inundation Depth and Flow Velocities Based on Runup. The maximum inundation depths and flow velocities associated with the stages of tsunami flooding shall be determined in accordance with Section 6.6.2. Calculated flow velocity shall not be taken as less than 10 ft/s (3.0 m/s) and need not be taken as greater than the lesser of $1.5(gh_{\max})^{1/2}$ and 50 ft/s (15.2 m/s).

Where the maximum topographic elevation along the topographic transect between the shoreline and the inundation limit is greater than the runup elevation, one of the following methods shall be used:

1. The site-specific procedure of Section 6.7.6 shall be used to determine inundation depth and flow velocities at the site, subject to the above range of calculated velocities.
2. For determination of the inundation depth and flow velocity at the site, the procedure of Section 6.6.2, Energy Grade Line Analysis, shall be used, assuming a runup elevation and horizontal inundation limit that has at least 100% of the maximum topographic elevation along the topographic transect.

6.6.2 Energy Grade Line Analysis of Maximum Inundation Depths and Flow Velocities. The maximum velocity and maximum inundation depth along the ground elevation profile

up to the inundation limit shall be determined using the Energy Grade Line Analysis. The orientations of the topographic transect profiles used shall be determined considering the requirements of Section 6.8.6.1. The ground elevation along the transect, z_i , shall be represented as a series of linear sloped segments each with a Manning's coefficient consistent with the equivalent terrain macroroughness friction of that terrain segment. The Energy Grade Line Analysis shall be performed incrementally in accordance with Eq. (6.6-1) across the topographic transect in a stepwise procedure. Eq. (6.6-1) shall be applied across the topographic transect from the runup where the hydraulic head at the inundation limit, x_R , is zero, and the water elevation is equal to the runup, R , by calculating the change in hydraulic head at each increment of terrain segment toward the shoreline until the site of interest is reached, as shown in Fig. 6.6-1.

$$E_{g,i} = E_{g,i-1} + (\varphi_i + s_i)\Delta x_i \quad (6.6-1)$$

where

$E_{g,i}$ = Hydraulic head at point $i = h_i + u_i^2/2g = h_i(1 + 0.5F_{ri}^2)$;

h_i = Inundation depth at point i ;

u_i = Maximum flow velocity at point i ;

φ_i = Average ground slope between points i and $i - 1$;

F_{ri} = Froude number = $u/(gh)^{1/2}$ at point i ;

$\Delta x_i = x_{i-1} - x_i$, the increment of horizontal distance, which shall not be coarser than 100 ft (30.5 m) spacing;

x_i = Horizontal distance inland from NAVD 88 shoreline at point i ; and

s_i = Friction slope of the energy grade line between points i and $i - 1$, is calculated per Eq. (6.6-2).

$$s_i = (u_i)^2 / ((1.49/n)^2 h_i^{4/3}) = gF_{ri}^2 / ((1.49/n)^2 h_i^{1/3}) \quad (6.6-2)$$

$$s_i = (u_i)^2 / ((1.00/n)^2 h_i^{4/3}) = gF_{ri}^2 / ((1.00/n)^2 h_i^{1/3}) \quad (6.6-2.si)$$

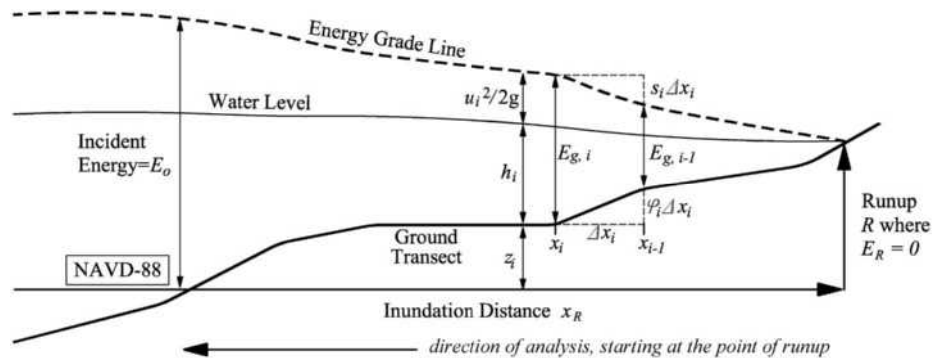
where

n = Manning's coefficient of the terrain segment being analyzed, according to Table 6.6-1, and

E_R = Hydraulic head of zero at the point of runup

Velocity shall be determined as a function of inundation depth, in accordance with the prescribed value of the Froude number calculated according to Eq. (6.6-3).

$$F_r = \alpha \left(1 - \frac{x}{x_R} \right)^{0.5} \quad (6.6-3)$$



Note: R = Design tsunami runup elevation above NAVD 88 datum; x_R = Design inundation distance inland from NAVD 88 shoreline; and z_i = Ground elevation above NAVD 88 datum at point i .

FIGURE 6.6-1 Energy Method for Overland Tsunami Inundation Depth and Velocity

Table 6.6-1 Manning's Roughness, n , for Energy Grade Line Analysis

Description of Frictional Surface	n
Coastal water nearshore bottom friction	0.025 to 0.03
Open land or field	0.025
All other cases	0.03
Buildings of at least urban density	0.04

where a value of the Froude number coefficient, α , of 1.0 shall be used. Where tsunami bores are required to be considered per Section 6.6.4, the tsunami bore conditions specified in Sections 6.10.2.3 and 6.10.3.3 shall be applied using the values of h_e and $(h_e u^2)_{\text{bore}}$ evaluated with $\alpha = 1.3$.

6.6.3 Terrain Roughness. It shall be permitted to perform inundation analysis assuming bare-earth conditions with equivalent macroroughness. Bed roughness shall be prescribed using the Manning's coefficient n . It shall be permitted to use the values listed in Table 6.6-1 or other values based on terrain analysis in the recognized literature or as specifically validated for the inundation model used.

6.6.4 Tsunami Bores. Tsunami bores shall be considered where any of the following conditions exist:

1. The prevailing nearshore bathymetric slope is 1/100 or milder,
2. Shallow fringing reefs or other similar step discontinuities in nearshore bathymetric slope occur,
3. Where historically documented,
4. As described in the recognized literature, or
5. As determined by a site-specific inundation analysis.

Where tsunami bores are deemed to occur, the tsunami bore conditions specified in Sections 6.10.2.3 and 6.10.3.3 shall be applied.

6.6.5 Amplified Flow Velocities. Flow velocities determined in this section shall be adjusted for flow amplification in accordance with Section 6.8.5 as applicable. The adjusted value need not exceed the maximum limit specified in Section 6.6.1.

6.7 INUNDATION DEPTHS AND FLOW VELOCITIES BASED ON SITE-SPECIFIC PROBABILISTIC TSUNAMI HAZARD ANALYSIS

When required by Section 6.5, the inundation depths and flow velocities shall be determined by site-specific inundation studies complying with the requirements of this section. Site-specific analyses shall use the ASCE Tsunami Design Geodatabase of geocoded reference points of Offshore Tsunami Amplitude and dominant waveform period shown in Fig. 6.7-1 as input to an inundation numerical model or shall use an integrated generation, propagation, and inundation model that replicates the given offshore tsunami waveform amplitude and period from the seismic sources given in Section 6.7.1. The ASCE Tsunami Design Geodatabase of geocoded reference points of Offshore 328-ft (100-m) depth, Tsunami Amplitude, H_T , and Predominant Period, T_{TSU} , of the Maximum Considered Tsunami is available at <http://asce7tsunami.online>.

6.7.1 Tsunami Waveform. The tsunami waveform used along the offshore boundary of 328 ft (100 m) bathymetry shall be

constructed in accordance with Eq. (6.7-1), as illustrated in Fig. 6.7-2.

$$\eta = a_1 e^{-[\omega(t-t_o)]^2} + a_2 e^{-[\omega(t-\frac{T_{\text{TSU}}}{2}-t_o)]^2} \quad (6.7-1)$$

where the total wave height of the waveform is $= \text{abs}(a_1) + \text{abs}(a_2)$, and

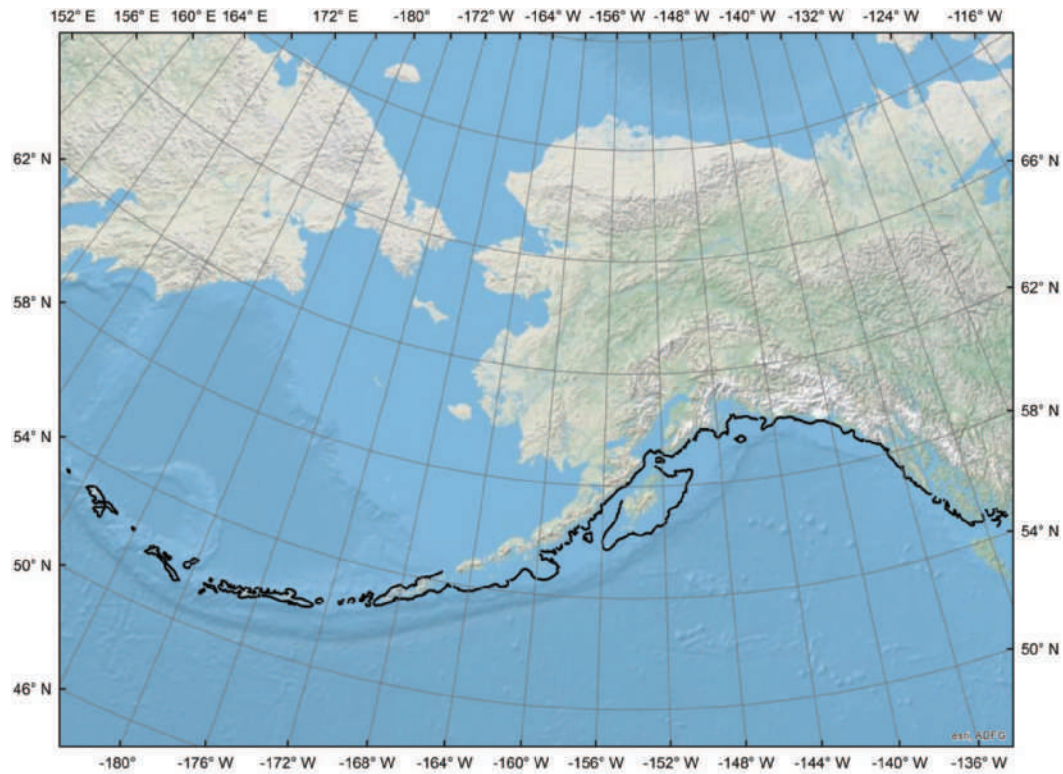
- η = the free surface elevation (in ft or m) as a function of time, t , used to drive the offshore boundary condition at the 328-ft (100-m) depth contour;
- a_1 = the amplitude of the leading pulse (in ft or m); it shall be negative for a leading depression tsunami;
- a_2 = the amplitude of the following, or second, pulse (in ft or m);
- T_{TSU} = wave period, or the time from the start of the first pulse to the end of the second pulse;
- ω = angular frequency of the waveform, equal to $2\pi/T_{\text{TSU}}$; and
- t_o = offset time of the wave train, generally set equal to T_{TSU}

The possibility of negative and positive leading amplitudes of the tsunami shall be considered, with the waveform given by Eq. (6.7-1) using the values of parameters given by the ASCE Tsunami Design Geodatabase of geocoded reference points shown in Fig. 6.7-1. For an inundation numerical model, the values given in Table 6.7-1 shall also be used to define at least two possible waveforms using the minimum and maximum prescribed values of a_2 . An integrated generation, propagation, and inundation model that replicates the given offshore tsunami waveform amplitude and period from the seismic sources need not use the values given in Table 6.7-1.

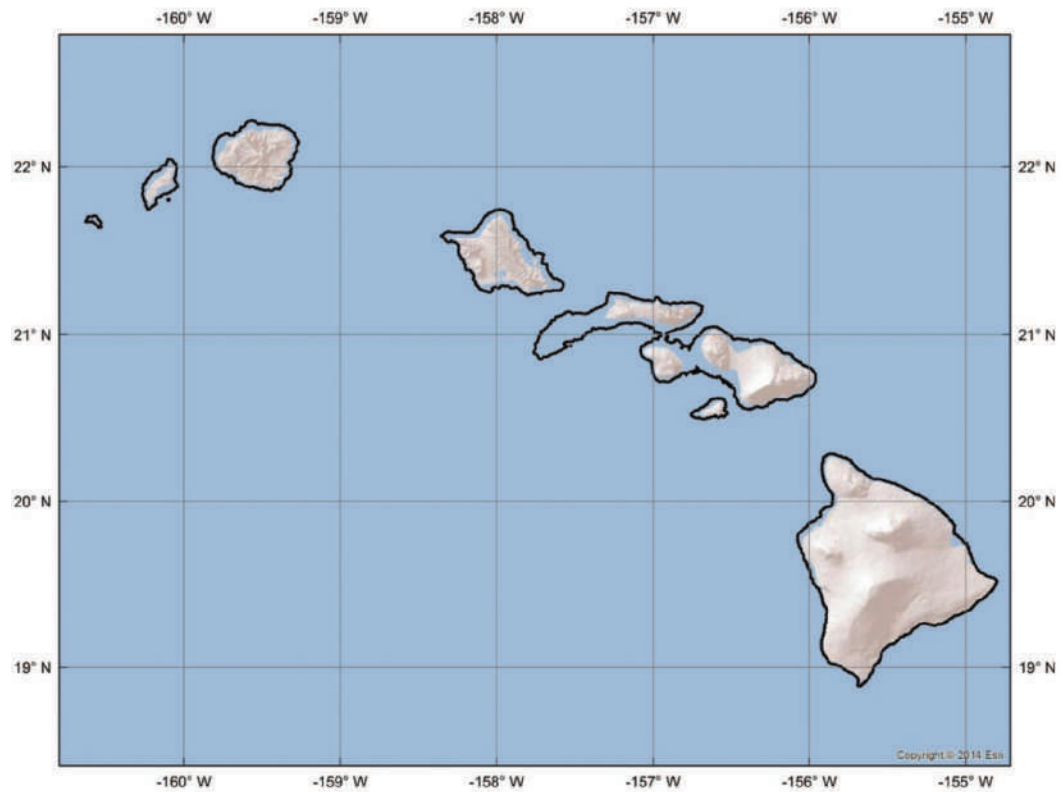
6.7.2 Tsunamigenic Sources. Tsunami sources shall consider the following to the extent that probabilistic hazards are documented in the recognized literature:

1. Local and distant subduction zone sources: It shall be permitted to use a system of delineated and discretized subduction zones in the Pacific basin comprised of systems of rectangular subfaults and their corresponding tectonic parameters.
 - a. Principal seismic sources shall include but are not restricted to Alaska: Alaska-Aleutian, Kamchatka-Kuril; California: Alaska-Aleutian, Cascadia, Kamchatka-Kuril, Chile-Peru; Hawaii: Alaska-Aleutian, Chile-Peru, Kamchatka-Kuril, Japan, Izu-Bonin-Mariana Islands; and Oregon and Washington: Cascadia, Alaska, Kamchatka-Kuril.
 - b. The maximum moment magnitude considered in the probability distribution of seismicity shall include the values given in Table 6.7-2.
2. Local, nonsubduction zone seismic fault sources capable of moment magnitude of 7 or greater, including offshore and/or submarine fault sources that are tsunamigenic.
3. Local coastal and submarine landslide sources documented in the recognized literature as being tsunamigenic of similar runup, as determined by historical evidence or having estimated probabilities within an order of magnitude of the principal seismic fault sources.

6.7.3 Earthquake Rupture Unit Source Tsunami Functions for Offshore Tsunami Amplitude. The tsunami modeling algorithm shall be based on earthquake rupture slip distributions for tsunami events, which shall be permitted to be represented by a linear combination of unit source functions



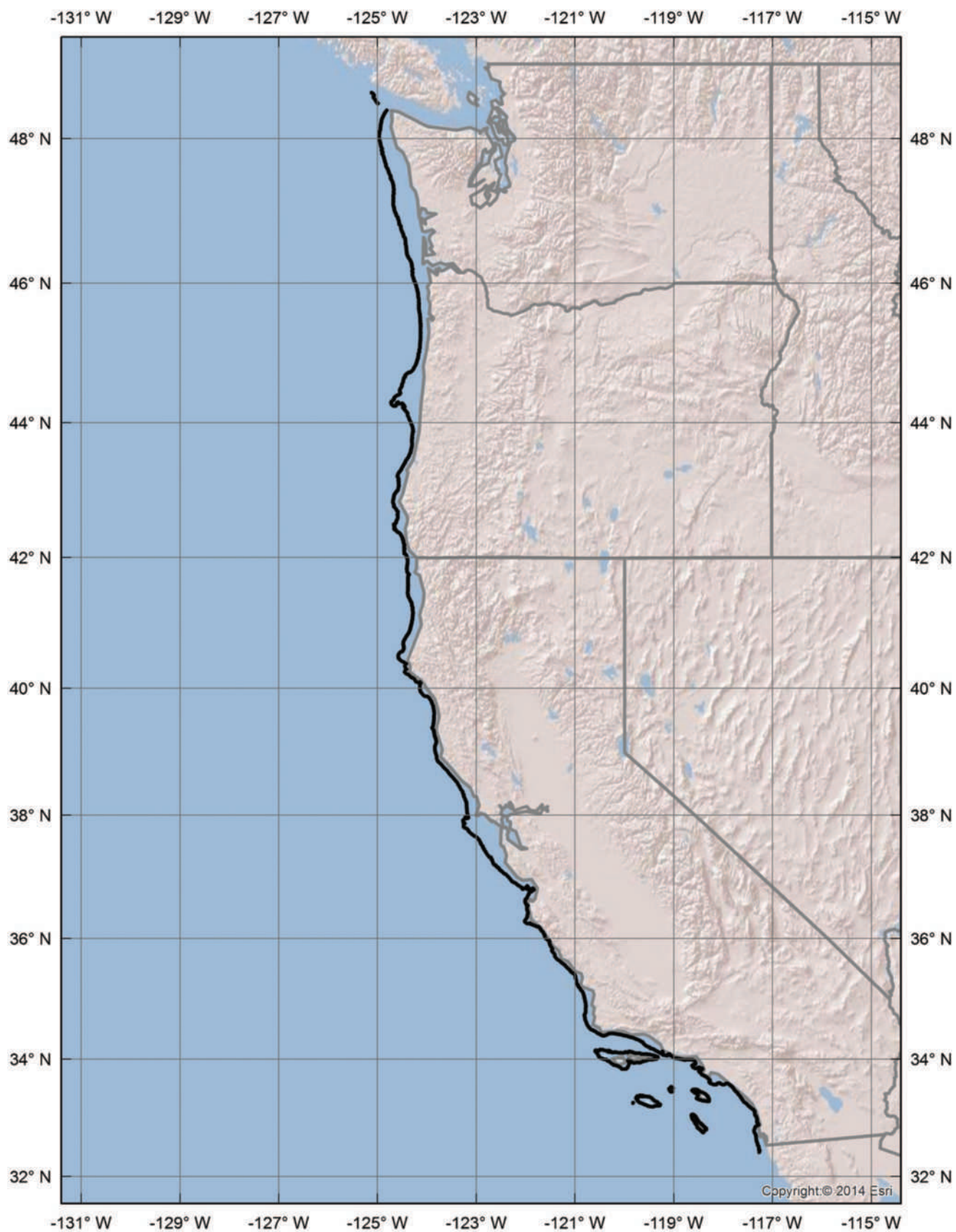
(a) Alaska



(c) Hawaii

Note: The ASCE Tsunami Design Geodatabase of geocoded points of Offshore (328-ft (100-m)) depth, Tsunami Amplitude, H_T , and Predominant Period, T_{TSU} , of the Maximum Considered Tsunami is available at <http://asce7tsunami.online>.

FIGURE 6.7-1 Extent of ASCE Tsunami Design Geodatabase of Geocoded Reference Points of Offshore (328-ft (100-m)) Depth, Tsunami Amplitude, H_T , and Predominant Period, T_{TSU} , of the Maximum Considered Tsunami (Continues)



(b) Pacific Coast

FIGURE 6.7-1 (Continued)

Table 6.7-1 Regional Waveform Parameters

Region	a_1	a_2^a	T_{TSU}^b
Washington	$1.0H_T$	-0.61 to $-0.82H_T$	30 to 40 minutes
Oregon	$1.0H_T$	-0.55 to $-0.67H_T$	30 to 45 minutes
Northern California	$1.0H_T$	-0.55 to $-0.67H_T$	25 to 35 minutes
Southern California	$1.0H_T$	-0.43 to $-0.67H_T$	25 to 35 minutes
Alaska	$1.0H_T$	-0.55 to $-0.82H_T$	20 to 40 minutes
Hawaii	$1.0H_T$	-0.67 to $-1.0H_T$	25 to 30 minutes

^aFor a leading depression waveform, the trough of amplitude a_2 shall precede the crest of amplitude a_1 .

^bThe value of T_{TSU} shall be used if no mapped value is given in Fig. 6.7-1.

Table 6.7-2 Maximum Moment Magnitude

Subduction Zone	Moment Magnitude $M_{w_{max}}$
Alaska-Aleutian	9.5
Cascadia	9.2
Chile-Peru	9.5
Izu-Bonin-Marianas	9.0
Kamchatka-Kurile and Japan Trench	9.4

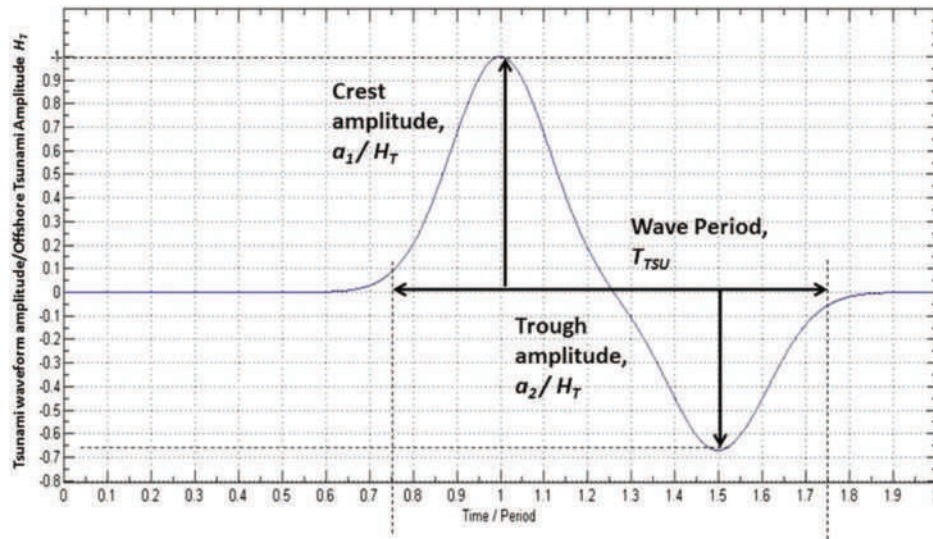


FIGURE 6.7-2 Illustration of Tsunami Offshore Incident Waveform Parameters at 328-ft (100-m) Depth

using a precomputed database of tsunami Green's source functions.

1. Tsunami waveform generation shall be permitted to be modeled by deconstructing a tsunami that is generated by an earthquake into a sum of individual tsunami waveforms composed from a scaled set of unit source subfaults that describe the earthquake rupture in terms of location, orientation, and rupture direction and sequence.
2. The waveforms defining the time series of wave height and velocity from a unit slip on each subfault shall be weighted by the actual slip or rupture distribution for the event and then summed linearly.
3. The algorithm shall account for coseismic vertical displacement.

6.7.4 Treatment of Modeling and Natural Uncertainties. A statistically weighted logic tree approach shall be used to account for epistemic uncertainties in the model parameters and shall provide a sample of tsunamigenic earthquakes and their occurrence probabilities from tectonic, geodetic, historical, and paleotsunami data, and estimated plate convergence rates, as follows:

1. Subdivide the occurrence probability systematically to account for variations in the parameters of magnitude, fault depth and geometry, and location, slip distribution,

and rupture extent of events consistent with maximum magnitudes, and tidal variation considering at least the Reference Sea Level.

2. To the extent practical and quantifiable, follow a similar logic tree approach to determine samples of tsunami sources such as nonsubduction zone earthquakes, landslides, and volcanic eruptions.

Aleatory uncertainties, such as the natural variability in the source processes, modeling uncertainties, and tidal variation as they relate to nearshore processes and wave runup, shall be included in the probabilistic analysis. When accounting for long wave durations with multiple maxima in the tsunami time series, it shall be permitted to consider tidal variability by selecting a rational tidal elevation independently from a probabilistic distribution of tide stages for each wave maximum. Truncation of aleatory distributions shall be chosen at an appropriate level for the return period but shall not be less than one standard deviation based on a regression analysis of computed versus observed data of Section 6.7.6.7.2.

6.7.5 Offshore Tsunami Amplitude. The probabilistic analysis shall be performed either by direct computation according to Section 6.7.5.2 or by performing a Probabilistic Tsunami Hazard Analysis for a region of interest to produce site-specific Offshore Tsunami Amplitude hazard maps and predominant wave period at 328-ft (100-m) depth in accordance with the following:

1. A Digital Elevation Model (DEM) from global, regional, and coastal data sets shall be used to cover the computational domain from the tsunami sources to the site. The bathymetry grid of the ocean shall have a DEM resolution finer than 4.35 mi (7,000 m), and the offshore model regime with depth greater than 656 ft (200 m) shall have a DEM resolution finer than 3,281 ft (1,000 m).
2. The Earth surface deformation shall be determined from the seismic source parameters using a planar fault model accounting for vertical changes to the seafloor.

6.7.5.1 Offshore Tsunami Amplitude for Distant Seismic Sources.

Offshore Tsunami Amplitude shall be probabilistically determined in accordance with the following:

1. A weighted combination of tsunami waveforms determined for each unit fault segment in accordance with the slip distribution shall be used to propagate tsunamis in deep water using the linear long wave equations, also termed the shallow water wave equations, where water depth is much less than the wavelength, to take into account spatial variations in seafloor depth.
2. The offshore wave amplitude distribution and associated wave parameters, including period, shall be determined for the design exceedance rate of the 2,475-year Maximum Considered Tsunami taking into consideration uncertainties per Section 6.7.4.
3. The analysis shall include the disaggregation of the seismic sources and associated moment magnitudes that together contribute at least 90% to the net offshore tsunami hazard at the site under consideration.
4. The value of offshore tsunami wave amplitude shall be not less than 80% of the value given by Fig. 6.7-1 for the Maximum Considered Tsunami offshore amplitude.

6.7.5.2 Direct Computation of Probabilistic Inundation and Runup.

It shall be permitted to compute probabilistic inundation and runup directly from a probabilistic set of sources, source characterizations, and uncertainties consistent with Section 6.7.2, Section 6.7.4, and the computing conditions set out in Section 6.7.6. The offshore wave amplitudes computed shall not be lower than 80% of the wave amplitudes given in Fig. 6.7-1.

6.7.6 Procedures for Determining Tsunami Inundation and Runup.

6.7.6.1 Representative Design Inundation Parameters.

Each disaggregated tsunami event shall be analyzed to determine representative design parameters consisting of maxima of runup, inundation depth, flow velocity, and momentum flux.

6.7.6.2 Seismic Subsidence before Tsunami Arrival.

Where the seismic source is a local subduction event, the Maximum Considered Tsunami inundation shall be determined for an overall elevation subsidence value shown in Fig. 6.7-3(a) and 6.7-3(b) or shall be directly computed for the seismic source mechanism. The GIS digital map layers of subsidence are available in the ASCE Tsunami Design Geodatabase at <http://asce7tsunami.online>.

6.7.6.3 Model Macroroughness Parameter.

It shall be permitted to perform inundation mapping under bare-earth conditions with macroroughness. Bed roughness shall be permitted to be prescribed using the Manning's coefficient n . Unless otherwise determined for the site, a default value of 0.025 or 0.030 shall be used for the ocean bottom and on land. Use of other values based on terrain analysis shall be justified in the

recognized literature or shall be specifically validated for the inundation model for field benchmarks of historical tsunami. Where values other than the defaults are used, the effects of degradation of roughness because of damaging flow characteristics shall be considered in the choice of Manning coefficient.

6.7.6.4 Nonlinear Modeling of Inundation.

Nonlinear shallow water wave equations or equivalent modeling techniques shall be used to transform the offshore wave amplitude from 328-ft (100-m) depth toward the shore to its nearshore tsunami amplitude and maximum inundation. The following effects shall be included as applicable to the bathymetry:

1. Shoaling, refraction, and diffraction to determine nearshore tsunami amplitude;
2. Dispersion effects in the case of short-wavelength sources, such as landslides and volcanic sources;
3. Reflected waves;
4. Channeling in bays;
5. Edge waves, and shelf and bay resonances;
6. Bore formation and propagation; and
7. Harbor and port breakwaters and levees.

6.7.6.5 Model Spatial Resolution.

A Digital Elevation Model (DEM) for the nearshore bathymetry depth of less than 656 ft (200 m) shall have a resolution not coarser than 295 ft (90 m). At bathymetric depths of less than 32.8 ft (10 m) and on land, the DEM shall have a resolution not coarser than the highest resolution available from the Tsunami Inundation Digital Elevation Models of the NOAA National Geophysical Data Center (NGDC). If a nested grid approach is used, the reduction in grid-spacing between consecutive grids shall not be more than a factor of 5. Where the NOAA NGDC models are not available, use of the best available USGS integrated Digital Elevation Model data shall be permitted when approved by the Authority Having Jurisdiction.

6.7.6.6 Built Environment.

If buildings and other structures are included for the purposes of more detailed flow analysis, the Digital Elevation Model resolution shall have a minimum resolution of 10 ft (3.0 m) to capture flow deceleration and acceleration in the built environment.

6.7.6.7 Inundation Model Validation

6.7.6.7.1 Historical or Paleotsunami Inundation Data.

Model scenario results shall be validated with available historical and/or paleotsunami records.

6.7.6.7.2 Model Validation by Benchmark Tests.

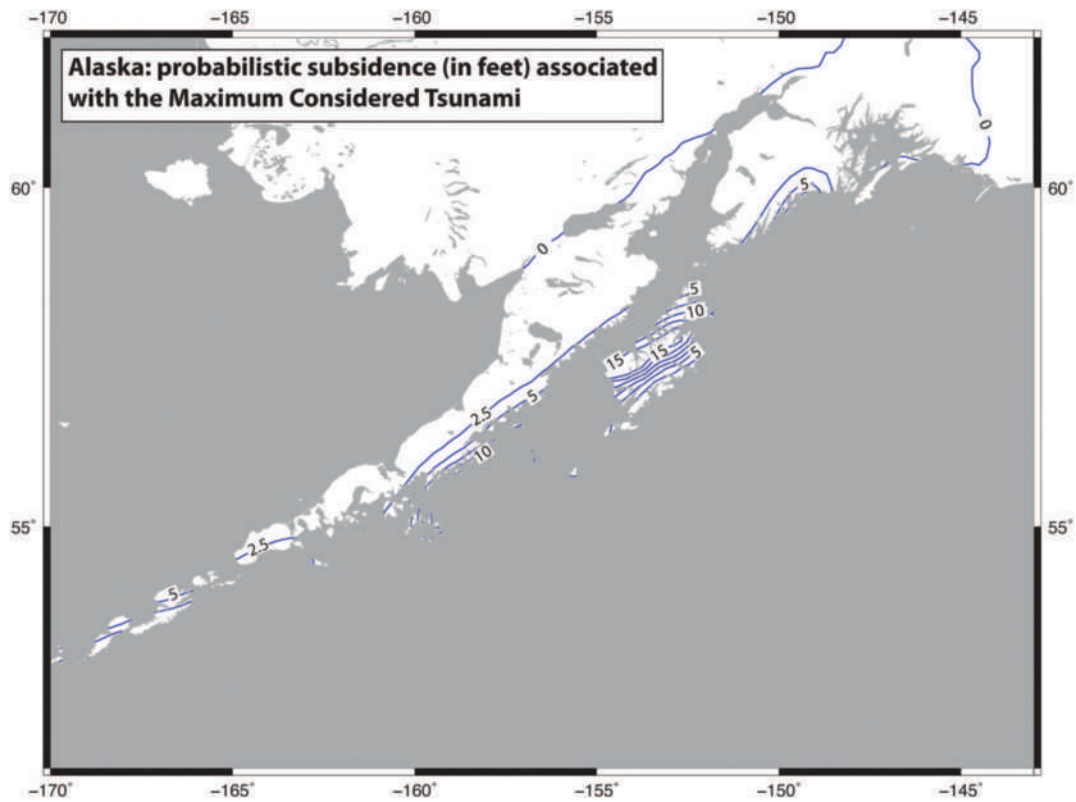
The inundation model shall be validated using the certification criteria of the National Tsunami Hazard Mitigation Program (NTHMP) within 10% by providing satisfactory performance in a series of benchmark tests of known data sets designated by the Tsunami Model Validation Advisory Group in NOAA Technical Memorandum OAR PMEL-135, Standards, Criteria, and Procedures for NOAA Evaluation of Tsunami Numerical Models, as modified by the NTHMP.

6.7.6.7.3 Tsunami Bore Formation or Soliton Fission.

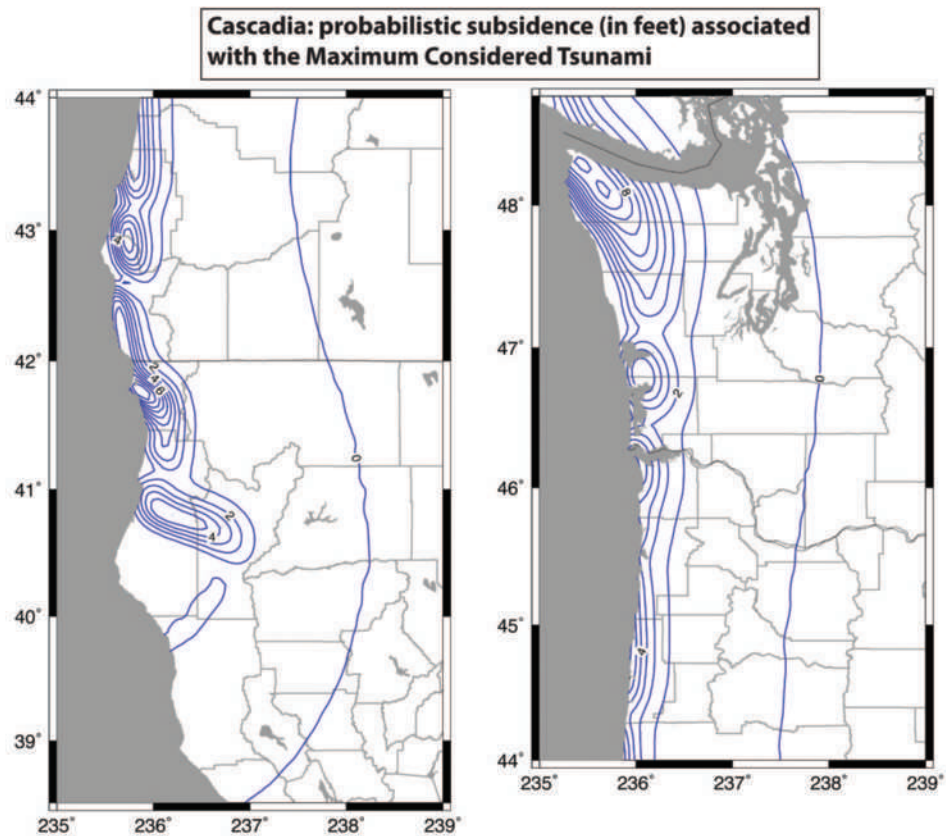
In regions where bore formation may occur, the model shall be validated with an independent scenario in the recognized literature, and acceptability shall be determined using modeled runup.

6.7.6.8 Determining Site-Specific Inundation Flow Parameters.

Inundation parameters for the scenarios from each disaggregated source region shall be determined. Probabilistic flow parameters shall be developed for the site



(a) Alaska



(b) Washington, Oregon, and Northern California

Note: The GIS digital map layers of subsidence are available from the ASCE Tsunami Design Geodatabase at <http://asce7tsunami.online>.

FIGURE 6.7-3 Earthquake-Induced Regional Ground Subsidence (in feet) associated with a Maximum Considered Tsunami Caused by a Local Subduction Earthquake

from the sample of computed tsunamis and their occurrence probabilities. Each tsunami event shall be analyzed to determine representative parameters such as maximum runup, inundation depth, flow velocity, and/or specific momentum flux by either of the following techniques:

1. Taking the weighted average of the scenario runs that bracket the offshore wave amplitude for the return period to determine hazard-consistent tsunami scenarios. The inundation limit shall be determined by the area that is inundated by the hazard-consistent tsunami scenario waves from the disaggregated principal seismic source zones affecting that site corresponding to the Maximum Considered Tsunami return period.
2. Determining the probabilistic distributions of flow parameters from the sample of computed tsunamis and their occurrence probabilities and reconstructing statistical distributions of flow parameters of inundation depth, velocity, and specific momentum flux at the site from the computed scenarios for at least three Load Cases, as indicated in Section 6.8.10.

In urban environments, the resulting flow velocities at a given structure location shall not be reduced from 90% of those determined in accordance with Section 6.6 before any velocity adjustments caused by flow amplification. For other terrain roughness conditions, the resulting flow velocities at a given structure location shall not be taken as less than 75% of those determined in accordance with Section 6.6 before any velocity adjustments caused by flow amplification.

6.7.6.9 Tsunami Design Parameters for Flow over Land. The flow parameters of inundation depth, flow velocity, and/or specific momentum flux at the site of interest shall be captured from a time history inundation analysis. Tsunami inundation depth and velocity shall be evaluated for the site at the stages of inundation defined by the Load Cases in Section 6.8.3.1. If the maximum momentum flux is found to occur at an inundation depth different than Load Case 2, the flow conditions corresponding to the maximum momentum flux shall be considered in addition to the Load Cases defined in Section 6.8.3.1.

6.8 STRUCTURAL DESIGN PROCEDURES FOR TSUNAMI EFFECTS

Structures, components, and foundations shall conform to the requirements of this section when subjected to the loads and effects of the Maximum Considered Tsunami.

6.8.1 Performance of Tsunami Risk Category II and III Buildings and Other Structures. Structural components, connections, and foundations of Risk Category II buildings and Risk Category III buildings and other structures shall be designed to meet Collapse Prevention Structural Performance criteria or better.

6.8.2 Performance of Tsunami Risk Category III Critical Facilities and Tsunami Risk Category IV Buildings and Other Structures. Tsunami Risk Category III Critical Facilities and Tsunami Risk Category IV buildings and other structures located within the Tsunami Design Zone shall be designed in accordance with the following requirements.

1. The operational nonstructural components and equipment of the building necessary for essential functions and the elevation of the bottom of the lowest horizontal structural member at the level supporting such components and equipment shall be above the inundation elevation of the Maximum Considered Tsunami.

2. Structural components and connections in occupiable levels and foundations shall be designed in accordance with Immediate Occupancy Structural Performance criteria. Occupiable levels shall be permitted where the elevation equals or exceeds the Maximum Considered Tsunami inundation elevation.
3. Tsunami Vertical Evacuation Refuge Structures shall also comply with Section 6.14.

6.8.3 Structural Performance Evaluation. Strength and stability shall be evaluated to determine that the design of the structure is capable of resisting the tsunami at the Load Cases defined in Section 6.8.3.1. The structural acceptance criteria for this evaluation shall be in accordance with either Section 6.8.3.4 or 6.8.3.5.

6.8.3.1 Load Cases. As a minimum, the following three Inundation Load Cases shall be evaluated:

1. Load Case 1: At an exterior inundation depth not exceeding the maximum inundation depth nor the lesser of one story or the height of the top of the first-story windows, the minimum condition of combined hydrodynamic force with buoyant force shall be evaluated with respect to the depth of water in the interior. The interior water depth shall be evaluated in accordance with Section 6.9.1.

EXCEPTION: Load Case 1 need not be applied to Open Structures nor to structures where the soil properties or foundation and structural design prevent detrimental hydrostatic pressurization on the underside of the foundation and lowest structural slab.

2. Load Case 2: Depth at two-thirds of maximum inundation depth when the maximum velocity and maximum specific momentum flux shall be assumed to occur in either incoming or receding directions.
3. Load Case 3: Maximum inundation depth when velocity shall be assumed at one-third of maximum in either incoming or receding directions.

The inundation depths and velocities defined for Load Cases 2 and 3 shall be determined by Fig. 6.8-1, unless a site-specific tsunami analysis is performed in accordance with Section 6.7.

6.8.3.2 Tsunami Importance Factors. The Tsunami Importance Factors, I_{TSU} , given in Table 6.8-1 shall be applied to the tsunami hydrodynamic and impact loads in Sections 6.10 and 6.11, respectively.

6.8.3.3 Load Combinations. Principal tsunami forces and effects shall be combined with other specified loads in accordance with the load combinations of Eq. (6.8-1):

$$0.9D + F_{TSU} + H_{TSU} \quad (6.8-1a)$$

$$1.2D + F_{TSU} + 0.5L + 0.2S + H_{TSU} \quad (6.8-1b)$$

where

F_{TSU} = tsunami load effect for incoming and receding directions of flow, and

H_{TSU} = load caused by tsunami-induced lateral foundation pressures developed under submerged conditions. Where the net effect of H_{TSU} counteracts the principal load effect, the load factor for H_{TSU} shall be 0.9

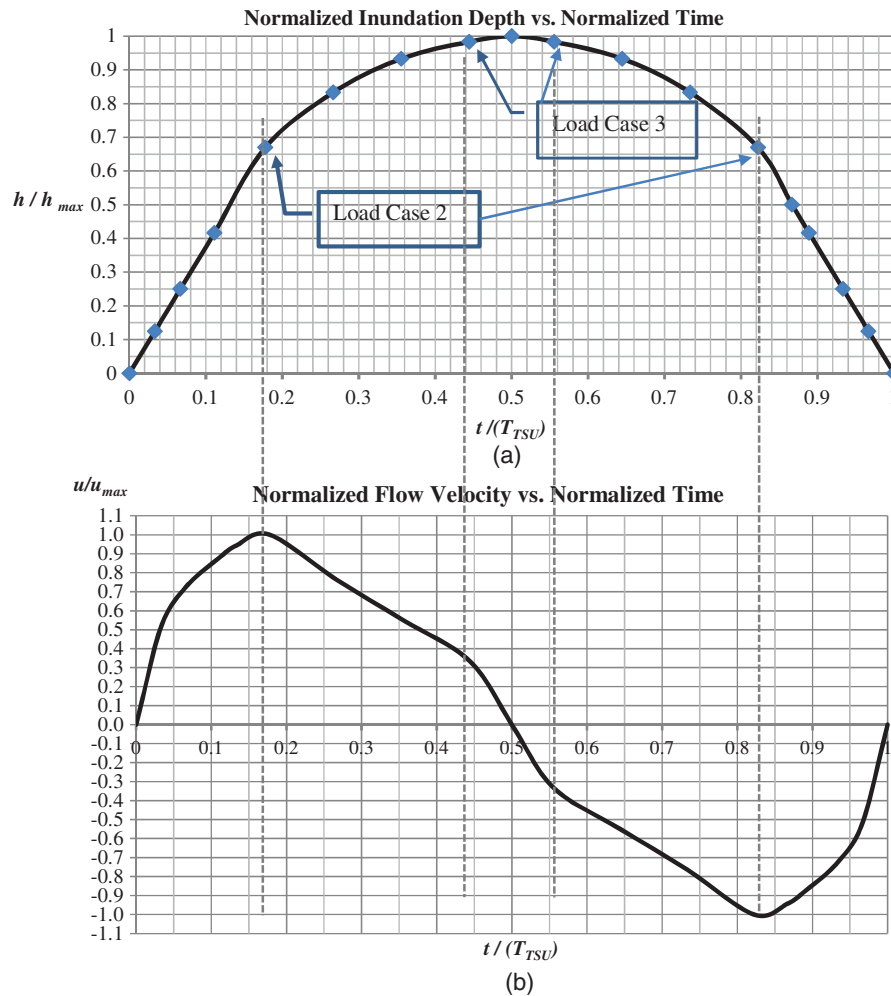


FIGURE 6.8-1 Inundation Load Cases 2 and 3

Table 6.8-1 Tsunami Importance Factors for Hydrodynamic and Impact Loads

Tsunami Risk Category	I_{tsu}
II	1.0
III	1.25
Tsunami Risk Category IV, Vertical Evacuation Refuges, and Tsunami Risk Category III Critical Facilities	1.25

6.8.3.4 Lateral-Force-Resisting System Acceptance Criteria. To evaluate the capacity of the structural system at the Life Safety Structural Performance Level to resist the lateral force effects of the design tsunami event for Seismic Design Category D, E, or F, it is permitted to use the value of 0.75 times the required Horizontal Seismic Load Effect, E_{mh} , which includes the system's overstrength factor, Ω_0 , as defined in Chapter 12 of this standard. For Immediate Occupancy Structural Performance objectives, the lateral-force-resisting system shall be explicitly analyzed and evaluated.

6.8.3.5 Structural Component Acceptance Criteria. Structural components shall be designed for the forces that result from the overall tsunami forces on the structural system combined with

any resultant actions caused by the tsunami pressures acting locally on the individual structural components for that direction of flow. Acceptance criteria of structural components shall be in accordance with Section 6.8.3.5.1, or in accordance with alternative procedures of 6.8.3.5.2 or 6.8.3.5.3, as applicable.

6.8.3.5.1 Acceptability Criteria by Component Design Strength. Internal forces and system displacements shall be determined using a linearly elastic, static analysis. The structural performance criteria required in Section 6.8.1, Section 6.8.2, and Section 6.8.3, as applicable, shall be deemed to comply if the design strength of the structural components and connections are shown to be greater than the Maximum Considered Tsunami loads and effects computed in accordance with the load combinations of Section 6.8.3.3. Material resistance factors, ϕ , shall be used as prescribed in the material-specific standards for the component and behavior under consideration.

6.8.3.5.2 Alternative Performance-Based Criteria

6.8.3.5.2.1 Alternative Analysis Procedures. It shall be permitted to use either a linear or nonlinear static analysis procedure. In a linear static analysis procedure, buildings and structures shall be modeled using an equivalent effective stiffness consistent with the secant value at or near the yield point. For a nonlinear static analysis procedure, a mathematical model directly incorporating

the nonlinear load-deformation characteristics of individual components of the structure shall be subjected to monotonically increasing loads until the required tsunami forces and applied actions are reached. For nonlinear static analysis procedures, expected deformation capacities shall be greater than or equal to the maximum deformation demands calculated at the required tsunami forces and applied actions. For debris impacts, it shall be permitted to use a nonlinear dynamic analysis procedure. For Tsunami Risk Category IV buildings and structures, an independent peer review shall be conducted as part of a review of the performance-based design by the Authority Having Jurisdiction.

6.8.3.5.2.2 **Alternative Structural Component Acceptability Criteria.** All actions shall be classified as either ductility-governed actions or force-sustained actions based on component inelastic behavior and the duration of the load effect, as follows:

1. Fluid forces in primary and secondary structural components detailed in accordance with the requirements of Seismic Design Category D, E, or F shall be evaluated as force-sustained actions.
2. Debris impacts and foundation settlement effects on primary and secondary structural components shall be evaluated as ductility-governed actions.
3. Debris impacts and foundation settlement effects on primary and secondary structural components not detailed in accordance with Seismic Design Category D, E, or F shall be evaluated as force-sustained actions.

For force-sustained actions, structural components shall have specified design strengths greater than or equal to the maximum design forces. Force-sustained actions shall be permitted to satisfy Eq. (6.8-2):

$$Q_{CS} \geq Q_{UF} \quad (6.8-2)$$

where

Q_{CS} = Specified strength of the structural element and
 Q_{UF} = Maximum force generated in the element because of gravity and tsunami loading.

Expected material properties as defined in ASCE 41 (2014) shall be permitted to be used for ductility-governed actions. Results of a linear analysis procedure shall not exceed the component acceptance criteria for linear procedures of ASCE 41 (2014), Chapters 9 through 11, for the applicable structural performance criteria required for the building or structure Tsunami Risk Category. Ductility-governed actions shall be permitted to satisfy Eq. (6.8-3):

$$mQ_{CE} \geq Q_{UD} \quad (6.8-3)$$

where

m = Value of the component demand modification factor defined in ASCE 41 (2014) to account for expected ductility at the required structural performance level;
 Q_{CE} = Expected strength of the structural element determined in accordance with ASCE 41 (2014); and
 Q_{UD} = Ductility-governed force caused by tsunami loading

6.8.3.5.3 **Alternative Acceptability by Progressive Collapse Avoidance.** Where tsunami loads or effects exceed acceptability criteria for a structural element or where required to accommodate extraordinary impact loads, it shall be permitted to check the

residual load-carrying capacity of the structure, assuming that the element has failed, in accordance with Section 2.5.2.2 and an alternate load path progressive collapse procedure in the recognized literature.

6.8.4 Minimum Fluid Density for Tsunami Loads. Seawater specific weight density γ_{sw} shall be taken as 64.0 lb/ft³ (10 kN/m³). Seawater mass density ρ_{sw} shall be taken as 2.0 slugs per cubic foot (sl/ft³) (1,025 kg/m³). The minimum fluid specific weight density γ_s for determining tsunami hydrostatic loads accounting for suspended solids and debris flow-embedded smaller objects shall be

$$\gamma_s = k_s \gamma_{sw} \quad (6.8-4)$$

The minimum fluid mass density, ρ_s , for determining tsunami hydrodynamic loads accounting for suspended solids and debris flow-embedded smaller objects shall be

$$\rho_s = k_s \rho_{sw} \quad (6.8-5)$$

where k_s , fluid density factor, shall be taken as 1.1.

6.8.5 Flow Velocity Amplification. The effect of upstream obstructing buildings and structures shall be permitted to be considered at a site that is exposed to the flow diffracting conditions given in Section 6.8.5.1 by any of the following:

1. A site-specific inundation analysis that includes modeling of the built environment in accordance with Section 6.7.6.6, or
2. The built environment is considered in the selection of Manning's roughness of Table 6.6-1 in accordance with the Energy Grade Line Analysis of Section 6.6.2, or
3. Site-specific physical or numerical modeling in accordance with Section 6.8.5.2 or Section 6.8.10, as applicable.

6.8.5.1 Upstream Obstructing Structures. The effect of upstream obstructions on flow shall be considered where the obstructions are enclosed structures of concrete, masonry, or structural steel construction located within 500 ft (152 m) of the site, and both of the following apply:

1. Structures have plan width greater than 100 ft (30.5 m) or 50% of the width of the downstream structure, whichever is greater.
2. The structures exist within the sector between 10 and 55 degrees to either side of the flow vector aligned with the center third of the width of the downstream structure.

6.8.5.2 Flow Velocity Amplification by Physical or Numerical Modeling. The effect of upstream structures on the flow velocity at a downstream site shall be permitted to be evaluated using site-specific numerical or physical modeling, as described in Section 6.7.6.6 or 6.8.10. The velocity determined for a "bare-earth" inundation shall be amplified for the conditions of Section 6.8.5.1. This analysis is not permitted to reduce the flow velocity except for structural countermeasures designed in accordance with Section 6.13.

6.8.6 Directionality of Flow

6.8.6.1 Flow Direction. Design of structures for tsunami loads and effects shall consider both incoming and outgoing flow conditions. The principal inflow direction shall be assumed to vary by ± 22.5 degrees from the transect perpendicular to the orientation of the shoreline averaged over 500 ft (152 m) to either side of the site. The center of rotation of the variation of transects

shall be located at the geometric center of the structure in plan at the grade plane.

6.8.6.2 Site-Specific Directionality. A site-specific inundation analysis performed in accordance with Section 6.7.6 shall be permitted to be used to determine directionality of flow, provided that the directionalities so determined shall be assumed to vary by at least ± 10 degrees.

6.8.7 Minimum Closure Ratio for Load Determination. Loads on buildings shall be calculated assuming a minimum closure ratio of 70% of the inundated projected area along the perimeter of the structure, unless it is an Open Structure as defined in Section 6.2. The load effect of debris accumulation against or within the Open Structure shall be considered by using a minimum closure ratio of 50% of the inundated projected area along the perimeter of the Open Structure. Open Structures need not be subject to Load Case 1 of Section 6.8.3.1.

6.8.8 Minimum Number of Tsunami Flow Cycles. Design shall consider a minimum of two tsunami inflow and outflow cycles, the first of which shall be based on an inundation depth at 80% of the Maximum Considered Tsunami (MCT), and the second of which shall be assumed to occur with the Maximum Considered Tsunami inundation depth at the site. Local scour effects determined in accordance with Section 6.12, caused by the first cycle, shall be assumed to occur at 80% of the MCT inundation depth at the site and shall be considered as an initial condition of the second cycle.

6.8.9 Seismic Effects on the Foundations Preceding Local Subduction Zone Maximum Considered Tsunami. Where designated in Fig. 6.7-3 as a site subject to a local subduction zone tsunami from an offshore subduction earthquake, the structure shall be designed for the preceding coseismic effects. The foundation of the structure shall be designed to resist the preceding earthquake ground motion and associated effects per Chapter 11 of this standard using the Maximum Considered Earthquake Geometric Mean (MCE_G) Peak Ground Acceleration of Figs. 22-7, 22-8, and 22-9. Building foundation design shall include changes in the site surface and the in situ soil properties resulting from the design seismic event as initial conditions for the subsequent design tsunami event. The geotechnical investigation report shall include evaluation of foundation effects in reference to seismic effects preceding the tsunami, consideration of slope instability, liquefaction, total and differential settlement, surface displacement caused by faulting, and seismically induced lateral spreading or lateral flow. The additional requirements of Section 6.12 shall also be evaluated.

6.8.10 Physical Modeling of Tsunami Flow, Loads, and Effects. Physical modeling of tsunami loads and effects shall be permitted as an alternative to the prescriptive procedures in Sections 6.8.5 (flow velocity amplification), 6.10 (hydrodynamic loads), 6.11 (debris impact loads), and 6.12 (foundation design), provided that it meets all the following criteria:

1. The facility or facilities used for physical modeling shall be capable of generating appropriately scaled flows and inundation depths as specified for Load Cases in Section 6.8.3.1.
2. The test facility shall be configured so that reflections and edge effects shall not significantly affect the test section during the duration of the experiments.
3. The scale factors used in the physical modeling shall not be less than those shown in Table 6.8-2. Scale model tests not

Table 6.8-2 Minimum Scale Factors for Physical Modeling

Model Element	Minimum Scale Factor
Individual buildings	1:25
Flow modeling for groups of buildings	1:200
Structural components (e.g., walls, columns, piers)	1:10
Geotechnical investigations	1:5

directly addressed in Table 6.8-2 shall include a justification of the model applicability and scaling procedures.

4. Debris impacts of full or partial components shall be tested at full scale unless accompanied by a justification of the appropriateness of scaled testing in terms of hydrodynamics and structural mechanics as well as material properties.
5. The report of test results shall include a discussion of the accuracy of load condition generation and scale effects caused by dynamic and kinematic considerations, including dynamic response of test structures and materials.
6. Test results shall be adjusted to account for effective density, as calculated in Section 6.8.4.
7. Test results shall be adjusted by the Importance Factor from Section 6.8.3.2.
8. Test results shall include the effects of flow directionality in accordance with Section 6.8.6. This inclusion can be accomplished either by direct testing of flow at varying angles of incidence or by a combination of numerical and physical modeling that takes into account directionality of flow.

6.9 HYDROSTATIC LOADS

6.9.1 Buoyancy. Reduced net weight caused by buoyancy shall be evaluated for all inundated structural and designated nonstructural elements of the building in accordance with Eq. (6.9-1). Uplift caused by buoyancy shall include enclosed spaces without tsunami breakaway walls that have opening area less than 25% of the inundated exterior wall area. Buoyancy shall also include the effect of air trapped below floors, including integral structural slabs, and in enclosed spaces where the walls are not designed to break away. All windows, except those designed for large missile wind-borne debris impact or blast loading, shall be permitted to be considered openings when the inundation depth reaches the top of the windows or the expected strength of the glazing, whichever is less. The volumetric displacement of foundation elements, excluding deep foundations, shall be included in this calculation of uplift.

$$F_v = \gamma_s V_w \quad (6.9-1)$$

6.9.2 Unbalanced Lateral Hydrostatic Force. Inundated structural walls with openings less than 10% of the wall area and either longer than 30 ft (9.14 m) without adjacent tsunami breakaway walls or having a two- or three-sided perimeter structural wall configuration regardless of length shall be designed to resist an unbalanced hydrostatic lateral force given by Eq. (6.9-2), occurring during the Load Case 1 and the Load Case 2 inflow cases defined in Section 6.8.3.1. In conditions where the flow overtops the wall, $h_{\max}$ in Eq. (6.9-2) is replaced with the height of the wall.

$$F_h = \frac{1}{2} \gamma_s b h_{\max}^2 \quad (6.9-2)$$

6.9.3 Residual Water Surge Load on Floors and Walls. All horizontal floors below the maximum inundation depth shall be designed for dead load plus a residual water surge pressure, p_r , given by Eq. (6.9-3). Structural walls that have the potential to retain water during drawdown shall also be designed for residual water hydrostatic pressure.

$$p_r = \gamma_s h_r \quad h_r = h_{\max} - h_s \quad (6.9-3)$$

where h_s = top of floor slab elevation. However, h_r need not exceed the height of the continuous portion of any perimeter structural element at the floor.

6.9.4 Hydrostatic Surge Pressure on Foundation. Hydrostatic surge pressure caused by tsunami inundation shall be calculated as

$$p_s = \gamma_s h_{\max} \quad (6.9-4)$$

6.10 HYDRODYNAMIC LOADS

Hydrodynamic loads shall be determined in accordance with this section. The structure's lateral-force-resisting system and all structural components below the inundation elevation at the site shall be designed for the hydrodynamic loads given in either Section 6.10.1 or 6.10.2. All wall and slab components shall also be designed for all applicable loads given in Section 6.10.3.

6.10.1 Simplified Equivalent Uniform Lateral Static Pressure. It shall be permitted to account for the combination of any unbalanced lateral hydrostatic and hydrodynamic loads by applying an equivalent maximum uniform pressure, p_{uw} , determined in accordance with Eq. (6.10-1), applied over 1.3 times the calculated maximum inundation depth $h_{\max}$ at the site, in each direction of flow.

$$p_{uw} = 1.25 I_{\text{tsu}} \gamma_s h_{\max} \quad (6.10-1)$$

6.10.2 Detailed Hydrodynamic Lateral Forces

6.10.2.1 Overall Drag Force on Buildings and Other Structures. The building lateral-force-resisting system shall be designed to resist overall drag forces at each level caused either by incoming or outgoing flow at Load Case 2 given by Eqs. (6.10-2) and (6.10-3).

$$F_{dx} = \frac{1}{2} \rho_s I_{\text{tsu}} C_d C_{cx} B (h u^2) \quad (6.10-2)$$

where C_d is the drag coefficient for the building as given in Table 6.10-1 and where C_{cx} is determined as

$$C_{cx} = \frac{\sum (A_{\text{col}} + A_{\text{wall}}) + 1.5 A_{\text{beam}}}{B h_{sx}} \quad (6.10-3)$$

and A_{col} and A_{wall} are the vertical projected areas of all individual column and wall elements. A_{beam} is the combined vertical projected area of the slab edge facing the flow and the deepest beam laterally exposed to the flow. The summation of these column, wall, and beam areas is divided by the overall building wall area of width B times the average of the story heights, h_{sx} , above and below each level for each story below the tsunami inundation height for each of the three Load Cases specified in Section 6.8.3.1. Any structural or nonstructural wall that is not a tsunami breakaway wall shall be included in the

Table 6.10-1 Drag Coefficients for Rectilinear Structures

Width to Inundation Depth ^a Ratio B/h_{sx}	Drag Coefficient C_d
<12	1.25
16	1.3
26	1.4
36	1.5
60	1.75
100	1.8
≥120	2.0

^aInundation depth for each of the three Load Cases of inundation specified in Section 6.8.3.1. Interpolation shall be used for intermediate values of width to inundation depth ratio B/h_{sx} .

Table 6.10-2 Drag Coefficients for Structural Components

Structural Element Section	Drag Coefficient C_d
Round column or equilateral polygon with six sides or more	1.2
Rectangular column of at least 2:1 aspect ratio with longer face oriented parallel to flow	1.6
Triangular pointing into flow	1.6
Freestanding wall submerged in flow	1.6
Square or rectangular column with longer face oriented perpendicular to flow	2.0
Triangular column pointing away from flow	2.0
Wall or flat plate, normal to flow	2.0
Diamond-shape column, pointed into the flow (based on face width, not projected width)	2.5
Rectangular beam, normal to flow	2.0
I, L, and channel shapes	2.0

$A_{\text{wall}} \cdot C_{cx}$ shall not be taken as less than the closure ratio value given in Section 6.8.7 but need not be taken as greater than 1.0.

6.10.2.2 Drag Force on Components. The lateral hydrodynamic load given by Eq. (6.10-4) shall be applied as a pressure resultant on the projected inundated height, h_e , of all structural components and exterior wall assemblies below the inundation depth

$$F_d = \frac{1}{2} \rho_s I_{\text{tsu}} C_d b (h_e u^2) \quad (6.10-4)$$

where for interior components the values of C_d given in Table 6.10-2 shall be used, and b is the component width perpendicular to the flow. For exterior components, a C_d value of 2.0 shall be used, and width dimension b shall be taken as the tributary width multiplied by the closure ratio value given in Section 6.8.7.

The drag force on component elements shall not be additive to the overall drag force computed in Section 6.10.2.1.

6.10.2.3 Tsunami Loads on Vertical Structural Components, F_w . The force F_w on vertical structural components shall be determined as the hydrodynamic drag forces in accordance with Eq. (6.10-5a). Where flow of a tsunami bore occurs with a Froude number at the site that is greater than 1.0 and where individual wall, wall pier, or column components have a width to inundation depth ratio of 3 or more, F_w shall be determined by Eq. (6.10-5b). Force F_w is applied to all vertical structural components that are wider than 3 times the

inundation depth corresponding to Load Case 2 during inflow as defined in Section 6.8.3.

$$F_w = \frac{1}{2} \rho_s I_{tsu} C_d b (h_e u^2) \quad (6.10-5a)$$

$$F_w = \frac{3}{4} \rho_s I_{tsu} C_d b (h_e u^2)_{bore} \quad (6.10-5b)$$

6.10.2.4 Hydrodynamic Load on Perforated Walls, F_{pw} . For walls with openings that allow flow to pass between wall piers, the force on the elements of the perforated wall F_{pw} shall be permitted to be determined using Eq. (6.10-6), but shall not be less than F_d per Eq. (6.10-4):

$$F_{pw} = (0.4C_{cx} + 0.6)F_w \quad (6.10-6)$$

6.10.2.5 Walls Angled to the Flow. For walls oriented at an angle less than 90° to the flow directions considered in Section 6.8.3, the transient lateral load per unit width, $F_{w\theta}$, shall be determined in accordance with Eq. (6.10-7).

$$F_{w\theta} = F_w \sin^2 \theta \quad (6.10-7)$$

where θ is the included angle between the wall and the direction of the flow.

6.10.3 Hydrodynamic Pressures Associated with Slabs

6.10.3.1 Flow Stagnation Pressure. The walls and slabs of spaces in buildings that are subject to flow stagnation pressurization shall be designed to resist pressure determined in accordance with Eq. (6.10-8).

$$P_p = \frac{1}{2} \rho_s I_{tsu} u^2 \quad (6.10-8)$$

where u is the maximum free flow velocity at that location and Load Case.

6.10.3.2 Hydrodynamic Surge Uplift at Horizontal Slabs. Slabs and other horizontal components shall be designed to resist the applicable uplift pressures given in this section.

6.10.3.2.1 Slabs Submerged during Tsunami Inflow. Horizontal slabs that become submerged during tsunami inundation inflow shall be designed for a minimum hydrodynamic uplift pressure of 20 psf (0.958 kPa) applied to the soffit of the slab. This uplift is an additional Load Case to any hydrostatic buoyancy effects required by Section 6.9.1.

6.10.3.2.2 Slabs over Sloping Grade. Horizontal slabs located over grade slope, ϕ , greater than 10 degrees shall be designed for a redirected uplift pressure applied to the soffit of the slab, given by Eq. (6.10-9), but not less than 20 psf (0.958 kPa).

$$P_u = 1.5 I_{tsu} \rho_s u_v^2 \quad (6.10-9)$$

where

$u_v = u \tan \phi$,

u = Horizontal flow velocity corresponding to a water depth equal to or greater than h_{ss} , the elevation of the soffit of the floor system, and

ϕ = Average slope of grade plane beneath the slab.

6.10.3.3 Tsunami Bore Flow Entrapped in Structural Wall-Slab Recesses. Hydrodynamic loads for bore flows entrapped in structural wall-slab recesses shall be determined in accordance with this section. The reductions of load given in Sections 6.10.3.3.2 to 6.10.3.3.5 may be combined multiplicatively, but the net load reduction shall not exceed the maximum individual reduction given by any one of these sections.

6.10.3.3.1 Pressure Load in Structural Wall-Slab Recesses. Where flow of a tsunami bore beneath an elevated slab is prevented by a structural wall located downstream of the upstream edge of the slab, the wall and the slab within h_s of the wall shall be designed for the outward pressure, P_u , of 350 psf (16.76 kPa). Beyond h_s , but within a distance of $h_s + l_w$ from the wall, the slab shall be designed for an upward pressure of half of P_u [i.e., 175 psf (8.38 kPa)]. The slab outside a distance of $h_s + l_w$ from the wall shall be designed for an upward pressure of 30 psf (1.436 kPa).

6.10.3.3.2 Reduction of Load with Inundation Depth. Where the inundation depth is less than two-thirds of the clear story height, the uplift pressures specified in Section 6.10.3.3.1 shall be permitted to be reduced in accordance with Eq. (6.10-10) but shall not be taken as less than 30 psf (1.436 kPa).

$$P_u = I_{tsu} \left(590 - 160 \frac{h_s}{h} \right) [\text{psf}] \quad (6.10-10)$$

$$P_u = I_{tsu} \left(28.25 - 7.66 \frac{h_s}{h} \right) [\text{kPa}] \quad (6.10-10si)$$

where h_s/h is the ratio of slab height to inundation depth.

6.10.3.3.3 Reduction of Load for Wall Openings. Where the wall blocking the bore below the slab has openings through which the flow can pass, the reduced pressure on the wall and slab shall be determined in accordance with Eq. (6.10-11).

$$P_{ur} = C_{cx} P_u \quad (6.10-11)$$

where C_{cx} is the ratio of the solid area of the wall to the total inundated area of the vertical plane of the inundated portion of the wall at that level.

6.10.3.3.4 Reduction in Load for Slab Openings. Where the slab is provided with an opening gap or breakaway panel designed to create a gap of width w_g , adjacent to the wall, then the uplift pressure on the remaining slab shall be determined in accordance with Eq. (6.10-12).

$$P_{ur} = C_{bs} P_u \quad (6.10-12)$$

$$\text{where for } w_g < 0.5h_s, \quad C_{bs} = 1 - \frac{w_g}{h_s} \quad (6.10-13)$$

$$\text{and for } w_g \geq 0.5h_s, \quad C_{bs} = 0.56 - 0.12 \frac{w_g}{h_s} \quad (6.10-14)$$

The value of C_{bs} shall not be taken as less than zero.

6.10.3.3.5 Reduction in Load for Tsunami Breakaway Wall. If the wall restricting the flow is designed as a tsunami breakaway wall, then the uplift on the slab shall be permitted to be

determined in accordance with Section 6.10.3.1, but it need not exceed the pressure equivalent to the total nominal shear force necessary to cause disengagement of the breakaway wall from the slab.

6.11 DEBRIS IMPACT LOADS

Debris impact loads shall be determined in accordance with this section. These loads need not be combined with other tsunami-related loads as determined in other sections of this chapter.

Where the minimum inundation depth is 3 ft (0.914 m) or greater, design shall include the effects of debris impact forces. The most severe effect of impact loads within the inundation depth shall be applied to the perimeter gravity-load-carrying structural components located on the principal structural axes perpendicular to the range of inflow or outflow directions defined in Section 6.8.7. Except as specified below, loads shall be applied at points critical for flexure and shear on all such members in the inundation depth being evaluated. Inundation depths and velocities corresponding to Load Cases 1, 2, and 3 defined in Section 6.8.3.1 shall be used. Impact loads need not be applied simultaneously to all affected structural components.

All buildings and other structures meeting the above requirement shall be designed for impact by floating wood poles, logs, and vehicles, and for tumbling boulders and concrete debris, per Sections 6.11.2 to 6.11.4. Where a site is proximate to a port or container yard, the potential for strikes from shipping containers and ships and barges shall be determined by the procedure in Section 6.11.5. Buildings and other structures determined by that procedure to lie in the hazard zone for strikes by shipping containers shall be designed for impact loads in accordance with Section 6.11.6. In lieu of Sections 6.11.2–6.11.6, it shall be permitted to alternatively evaluate the impacts by poles, logs, vehicles, tumbling boulders, concrete debris, and shipping containers by applying the alternative simplified static load of Section 6.11.1.

Tsunami Risk Category III Critical Facilities and Tsunami Risk Category IV buildings and structures determined to be in the hazard zone for strikes by ships and barges in excess of 88,000 lb (39,916 kg) Deadweight Tonnage (DWT), as determined by the procedure of Section 6.11.5, shall be designed for impact by these vessels in accordance with Section 6.11.7.

6.11.1 Alternative Simplified Debris Impact Static Load. It shall be permitted to account for debris impact by applying the force given by Eq. (6.11-1) as a maximum static load, in lieu of the loads defined in Sections 6.11.2 to 6.11.6. This force shall be applied at points critical for flexure and shear on all such members in the inundation depth corresponding to Load Case 3 defined in Section 6.8.3.1.

$$F_i = 330C_o I_{tsu} \text{ [kips]} \quad (6.11-1)$$

$$F_i = 1,470C_o I_{tsu} \text{ [kN]} \quad (6.11-1.si)$$

where C_o is the orientation coefficient, equal to 0.65.

Where it is determined by the site hazard assessment procedure of Section 6.11.5 that the site is not in an impact zone for shipping containers, ships, and barges, then it shall be permitted to reduce the simplified debris impact force to 50% of the value given by Eq. (6.11-1).

6.11.2 Wood Logs and Poles. The nominal maximum instantaneous debris impact force, F_{ni} , shall be determined in accordance with Eq. (6.11-2).

$$F_{ni} = u_{\max} \sqrt{k m_d} \quad (6.11-2)$$

The design instantaneous debris impact force, F_i , shall be determined in accordance with Eq. (6.11-3).

$$F_i = I_{tsu} C_o F_{ni} \quad (6.11-3)$$

where

I_{tsu} = Importance Factor (given in Table 6.8-1);

C_o = Orientation coefficient, equal to 0.65 for logs and poles;
 $u_{\max}$ = Maximum flow velocity at the site occurring at depths sufficient to float the debris;

k = Effective stiffness of the impacting debris or the lateral stiffness of the impacted structural element(s) deformed by the impact, whichever is less; and

m_d = Mass W_d/g of the debris.

Logs and poles are assumed to strike longitudinally for calculation of debris stiffness in Eq. (6.11-2). The stiffness of the log or pole shall be calculated as $k = EA/L$, in which E is the longitudinal modulus of elasticity of the log, A is its cross-sectional area, and L is its length. A minimum weight of 1,000 lb (454 kg) and minimum log stiffness of 350 kip/in. (61,300 kN/m) shall be assumed.

The impulse duration for elastic impact shall be calculated from Eq. (6.11-4):

$$t_d = \frac{2m_d u_{\max}}{F_{ni}} \quad (6.11-4)$$

For an equivalent elastic static analysis, the impact force shall be multiplied by the dynamic response factor $R_{\max}$ specified in Table 6.11-1. To obtain intermediate values of $R_{\max}$, linear interpolation shall be used. For a wall, the impact shall be assumed to act along the horizontal center of the wall, and the natural period shall be permitted to be determined based on the fundamental period of an equivalent column with width equal to one-half of the vertical span of the wall.

It also shall be allowed to use an alternative method of analysis per Section 6.11.8.

6.11.3 Impact by Vehicles. An impact of floating vehicles shall be applied to vertical structural element(s) at any point greater

Table 6.11-1 Dynamic Response Ratio for Impulsive Loads, $R_{\max}$

Ratio of Impact Duration to Natural Period of the Impacted Structural Element	$R_{\max}$ (Response Ratio)
0.0	0.0
0.1	0.4
0.2	0.8
0.3	1.1
0.4	1.4
0.5	1.5
0.6	1.7
0.7	1.8
0.9	1.8
1.0	1.7
1.1	1.7
1.2	1.6
1.3	1.6
≥1.4	1.5

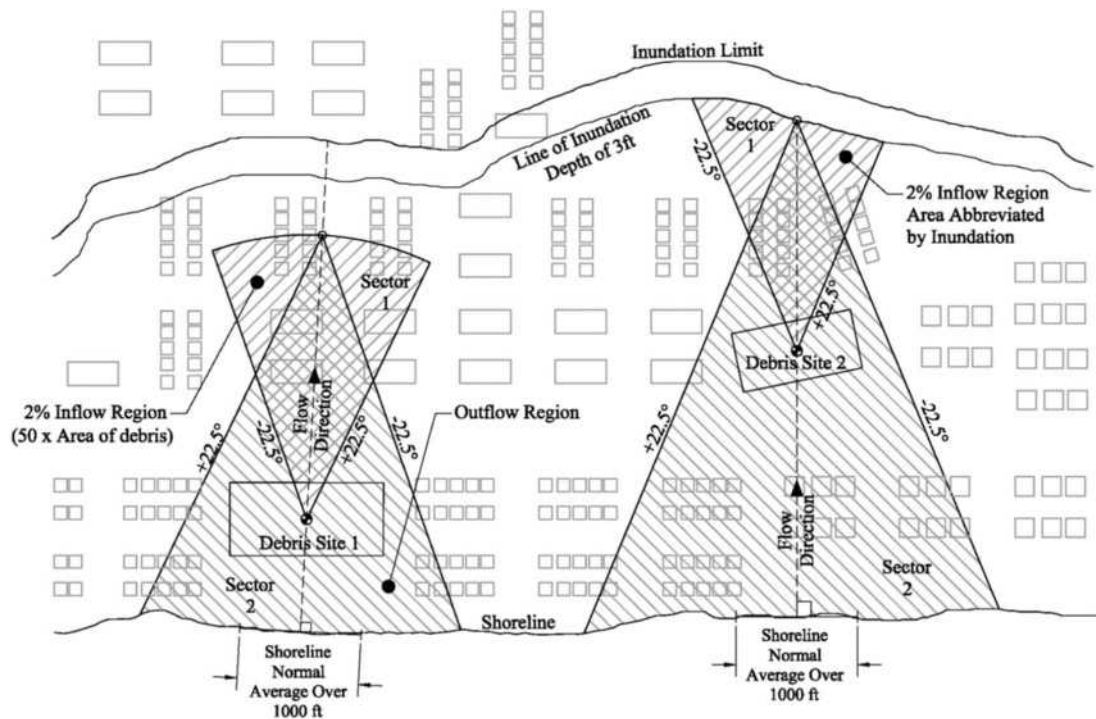


FIGURE 6.11-1 Illustration of Determination of Floating Debris Impact Hazard Region [1 ft = 0.3048 m]

than 3 ft (0.914 m) above grade up to the maximum depth. The impact force shall be taken as 30 kip (130 kN) multiplied by I_{tsu} .

6.11.4 Impact by Submerged Tumbling Boulder and Concrete Debris. Where the maximum inundation depth exceeds 6 ft (1.83 m), an impact force of 8,000 lb (36 kN) multiplied by I_{tsu} shall be applied to vertical structural element(s) at 2 ft (0.61 m) above grade.

6.11.5 Site Hazard Assessment for Shipping Containers, Ships, and Barges. Shipping containers and ships or barges disbursed from container yards, ports, and harbors shall be evaluated as potential debris impact objects. In such cases, a probable dispersion region shall be identified for each source to determine if the structure is located within a debris impact hazard region, as defined by the procedure in this section. If the structure is within the debris impact hazard region, then impact by shipping containers and/or ships and barges, as appropriate, shall be evaluated per Sections 6.11.6 and 6.11.7.

The expected total plan area of the debris objects at the source shall be determined. For containers, this is the average number of on-site containers multiplied by their plan area. For barges, the area of a nominal AASHTO (2009) design barge [195 × 35 ft, or 6,825 ft² (59.5 × 10.67 m, or 635 m²)] shall be multiplied by the average number of barges at the source. For ships, the average vessel deck plan area at the site shall be used. The geographic center of the source shall be identified, together with the primary flow direction, as defined in Section 6.8.6.1. Lines ±22.5° from this centerline shall be projected in the direction of tsunami inflow, as shown in Fig. 6.11-1. If topography (such as hills) will bound the water from this 45° sector, the direction of the sector shall be rotated to accommodate hill lines or the wedge shall be narrowed where it is constrained on two or more sides.

First, an arc of the debris impact hazard region for inflow shall be drawn as follows: one arc and the two radial boundary lines of

the 45° sector defines a circular sector region with an area that is 50 times the total sum debris area of the source, representing a 2% concentration of debris. However, the inland extent of the arc shall be permitted to be curtailed in accordance with any of the following boundaries:

- The extent of the sector shall be permitted to be curtailed where the maximum inundation depth is less than 3 ft (0.914 m), or in the case of ships where the inundation depth is less than the ballasted draft plus 2 ft (0.61 m).
- Structural steel and/or concrete structures shall be permitted to be considered to act as an effective grounding depth terminator of the sector if their height is at least equal to (1) for containers and barges, the inundation depth minus 2 ft (0.61 m), or (2) for ships, the inundation depth minus the sum of the ballasted draft and 2 ft (0.61 m).

Second, the debris impact hazard region for inflow and outflow shall be determined by rotating the circular segment by 180° and placing the center at the intersection of the centerline and the arc that defines the 2% concentration level or approved alternative boundary, as defined above. Buildings and other structures contained only in the first sector shall be designed for strikes by a container and/or other vessel carried with the inflow. Buildings and other structures contained only in the second sector shall be designed for strikes by a container and/or other vessel carried in the outflow. Buildings and other structures contained in both sectors shall be designed for strikes by a container and/or other vessel moving in either direction.

6.11.6 Shipping Containers. The impact force from shipping containers shall be calculated from Eqs. (6.11-2) and (6.11-3). The mass m_d is the mass of the empty shipping container. It shall be assumed that the strike contact is from one bottom corner of

Table 6.11-2 Weight and Stiffness of Shipping Container Waterborne Floating Debris

Type of Debris	Weight	Debris Stiffness (k)
20-ft (6.1-m) standard shipping container oriented longitudinally	Empty: 5,000 lb (2,270 kg) Loaded: 29,000 lb (13,150 kg)	245 kip/in. (42,900 kN/m)
40-ft (12.2-m) standard shipping container oriented longitudinally	Empty: 8,400 lb (3,810 kg) Loaded: 38,000 lb (17,240 kg)	170 kip/in. (29,800 kN/m)

the front (or rear) of the container. The container stiffness is $k = EA/L$, in which E is the modulus of elasticity of the bottom rail of the container, A is the cross-sectional area of the bottom rail, and L is the length of the bottom rail of the container. Minimum values are provided in Table 6.11-2. C_o , the orientation factor, shall be taken as equal to 0.65 for shipping containers.

The nominal design impact force, F_{ni} , from Eq. (6.11-2) for shipping containers need not be taken as greater than 220 kips (980 kN).

For empty shipping containers, the impulse duration for elastic impact shall be calculated from Eq. (6.11-4).

For loaded shipping containers the duration of the pulse is determined from Eq. (6.11-5):

$$t_d = \frac{(m_d + m_{\text{contents}})u_{\text{max}}}{F_{ni}} \quad (6.11-5)$$

in which m_{contents} shall be taken to be 50% of the maximum rated content capacity of the shipping container. Minimum values of $(m_d + m_{\text{contents}})$ are given in Table 6.11-2 for loaded shipping containers. The design shall consider both empty and loaded shipping containers.

For an equivalent static analysis, the impact force shall be multiplied by the dynamic response factor R_{max} specified in Table 6.11-1. To obtain intermediate values of R_{max} , linear interpolation shall be used. For a wall, the impact shall be assumed to act along the horizontal center of the wall, and the natural period shall be permitted to be determined based on the period of an equivalent column with width equal to one-half of the vertical span of the wall.

It also shall be permitted to use an alternative method of analysis per Section 6.11.8.

6.11.7 Extraordinary Debris Impacts. Where the maximum inundation depth exceeds 12 ft (3.66 m), extraordinary debris impacts of the largest deadweight tonnage vessel with ballasted draft less than the inundation depth within the debris hazard region of piers and wharves defined in Section 6.11.5 shall be assumed to impact the perimeter of Tsunami Risk Category III Critical Facilities and Tsunami Risk Category IV buildings and structures anywhere from the base of the structure up to 1.3 times the inundation depth plus the height to the deck of the vessel. The load shall be calculated from Eq. (6.11-3), based on the stiffness of the impacted structural element and a weight equal to the Lightship Weight (LWT) plus 30% of Deadweight Tonnage (DWT). An alternative analysis of Section 6.11.8 shall be permitted. Either as the primary approach, or where the impact loads exceed acceptability criteria for any structural element subject to impact, it is permitted to accommodate the impact through the alternative load path progressive collapse provisions of Section 6.8.3.5.3, applied to all framing levels from the base up to the story level above 1.3 times the inundation depth plus the height to the deck of the vessel as measured from the waterline.

6.11.8 Alternative Methods of Response Analysis. A dynamic analysis is permitted to be used to determine the structural response to the force applied as a rectangular pulse of duration time t_d with the magnitude calculated in accordance with Eq. (6.11-3). If the impact is large enough to cause inelastic behavior in the structure, it shall be permitted to use an equivalent single degree of freedom mass-spring system with a nonlinear stiffness that considers the ductility of the impacted structure for the dynamic analysis. Alternatively, for inelastic impact, the structural response shall be permitted to be calculated based on a work-energy method with nonlinear stiffness that incorporates the ductility of the impacted structure. The velocity applied in the work-energy method of analysis shall be u_{max} multiplied by the product of Importance Factor, I_{tsu} , and the orientation factor, C_o .

6.12 FOUNDATION DESIGN

Design of structure foundations and tsunami barriers shall provide resistance to the loads and effects of Section 6.12.2, shall provide capacity to support the structural load combinations defined in Section 6.8.3.1, and shall accommodate the displacements determined in accordance with Section 6.12.2.6. Foundation embedment depth and the capacity of the exposed piles to resist structural loads, including grade beam loads, shall both be determined taking into account the cumulative effects of general erosion and local scour. Alternatively, it shall be permitted to use the performance-based criteria of Section 6.12.3.

Site characterization shall include relevant information specified in Section 11.8, Geotechnical Investigation Report Requirements for Subsurface Soil Conditions.

6.12.1 Resistance Factors for Foundation Stability Analyses. The resistance factor of ϕ shall be assigned a value of 0.67 applied to the resisting capacities for use with stability analyses and for potential failures associated with bearing capacity, lateral pressure, internal stability of geotextile and reinforced earth systems, and slope stability, including drawdown conditions. A resistance factor of 0.67 shall also be assigned for the resisting capacities of uplift resisting anchorage elements.

6.12.2 Load and Effect Characterization. Foundations and tsunami barriers shall be designed to accommodate the effects of lateral earth pressure in accordance with Section 3.2, hydrostatic forces computed in accordance with Section 6.9, hydrodynamic loads computed in accordance with Section 6.10, and uplift and underseepage forces computed in accordance with Section 6.12.2.1. Foundations shall provide the capacity to withstand uplift and overturning from tsunami hydrostatic, hydrodynamic, and debris loads applied to the building superstructure. In addition, the effect of soil strength loss, general erosion, and scour shall be considered in accordance with the requirements of this section. A minimum of two wave cycles shall be considered for such effects.

6.12.2.1 Uplift and Underseepage Forces. Tsunami uplift and underseepage forces shall be evaluated as described in this section.

1. Uplift and underseepage forces shall include the three inundation Load Cases defined in Section 6.8.3.1.
2. Strength loss caused by scour and other soil effects such as liquefaction and pore pressure softening shall be considered. Additionally, uplift and underseepage forces on the foundation shall be determined for cases where
 - a. The soil is expected to be saturated before the tsunami, or
 - b. Soil saturation is anticipated to occur over the course of the incoming series of tsunami waves, or
 - c. The area of concern is expected to remain inundated after the tsunami.
3. The effect of live load and snow load shall not be used for uplift resistance.

6.12.2.2 Loss of Strength. Loss of shear strength because of tsunami-induced pore pressure softening shall be accounted for up to a depth of 1.2 times the maximum inundation depth, per Section 6.12.2.5. Tsunami-induced pore pressure softening need not be considered at locations where the maximum Froude number is less than 0.5.

6.12.2.3 General Erosion. General erosion during tsunami inundation runup and drawdown conditions shall be considered. Analysis of general erosion shall account for flow amplification as described in Section 6.8.5; it shall also account for enhancement caused by tsunami-induced pore pressure softening.

EXCEPTION: Analysis of general erosion is not required for rock or other nonerodible strata that are capable of preventing scour from tsunami flow of 30 ft/s (9.14 m/s).

General erosion during drawdown conditions shall consider flow concentration in channels, including channels newly formed during tsunami inundation and drawdown (channelized scour). Analysis of channelized scour need not include enhancement caused by pore pressure softening.

6.12.2.4 Scour. The depth and extent of scour shall be evaluated using the methods of Sections 6.12.2.4.1 and 6.12.2.4.2.

EXCEPTION: Scour evaluation is not required for rock or other nonerodible strata that prevent scour from tsunami flow of 30 ft/s (9.14 m/s) nor for Open Structures.

6.12.2.4.1 Sustained Flow Scour. Scour, including the effects of sustained flow around structures and including building corner piles, shall be considered. Sustained flow scour design depth and area extent shall be determined by dynamic numerical or physical modeling or empirical methods in the recognized literature. It shall be permitted to determine sustained flow scour and associated pore pressure softening in accordance with Table 6.12-1 and Fig. 6.12-1. Local scour depth caused by sustained flow given by Table 6.12-1 and Fig. 6.12-1 shall be permitted to be reduced by an adjustment factor in areas where the maximum flow Froude number is less than 0.5. The adjustment factor shall be taken as varying linearly from 0 at the horizontal inundation limit to 1.0 at the point where the Froude number is 0.5. The assumed area limits shall be considered to encompass the exposed building perimeter and to extend either side of the foundation perimeter a distance equal to the scour depth for consolidated or cohesive soils and a distance equal to three times the scour depth for nonconsolidated or noncohesive soils.

6.12.2.4.2 Plunging Scour. Plunging scour horizontal extent and depth shall be determined by dynamic numerical or physical

Table 6.12-1 Design Scour Depth Caused by Sustained Flow and Pore Pressure Softening

Inundation Depth h	Scour Depth D^a
<10 ft (3.05 m)	1.2 h
≥10 ft (3.05 m)	12 ft (3.66 m)

^aNot applicable to scour at sites with intact rock strata.

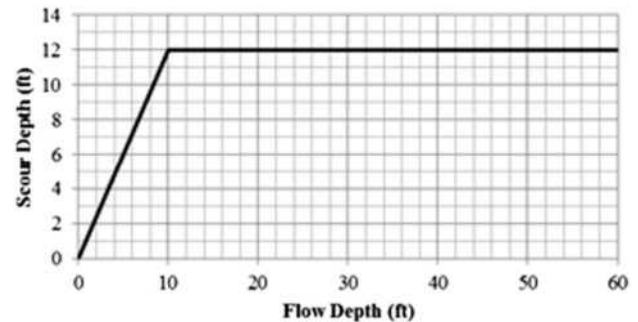


FIGURE 6.12-1 Scour Depth Caused by Sustained Flow and Pore Pressure Softening [1 ft = 0.305 m]

modeling or by empirical methods. In the absence of site-specific dynamic modeling and analysis, the plunging scour depth D_s shall be determined by Eq. (6.12-1).

$$D_s = c_{2V} \sqrt{\frac{qU \sin \psi}{g}} [\text{U.S. standard or SI units}] \quad (6.12-1)$$

where

c_{2V} = Dimensionless scour coefficient, permitted to be taken as equal to 2.8;

ψ = Angle between the jet at the scour hole and the horizontal, taken as the lesser value of 75° and the side slope of the overtopped structure on the scoured side, in the absence of other information;

g = Acceleration caused by gravity;

q = Discharge per unit width over the overtopped structure, as illustrated in Fig. 6.12-2 and calculated in accordance with Eq. (6.12-2); and

U = Jet velocity approaching the scour hole, obtained in accordance with Eq. (6.12-4).

$$q = C_{dis} \frac{2}{3} \sqrt{2g} H_o^{3/2} \quad (6.12-2)$$

where C_{dis} is a dimensionless discharge coefficient obtained in accordance with Eq. (6.12-3):

$$C_{dis} = 0.611 + 0.08 \frac{H_o}{H_B} \quad (6.12-3)$$

U is the jet velocity approaching the scour hole, resulting from the drop between the height h of the upstream water surface, plus any additional elevation difference d_d on the scouring side, in accordance with Eq. (6.12-4):

$$U = \sqrt{2g(h + d_d)} \quad (6.12-4)$$

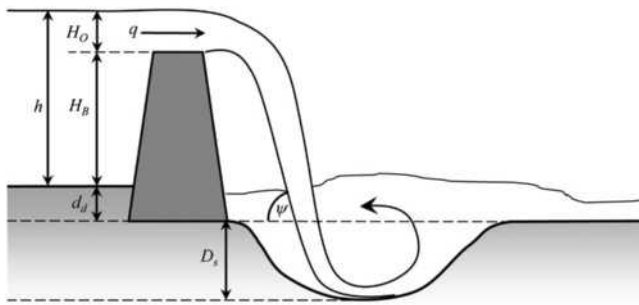


FIGURE 6.12-2 Plunging Scour Parameters

where d_d is the additional elevation difference between the upstream and scouring sides of the structure, as illustrated in Fig. 6.12-2.

6.12.2.5 Horizontal Soil Loads. Horizontal soil loads caused by unbalanced scour shall be included in the design of foundation elements.

6.12.2.6 Displacements. Vertical and horizontal displacements of foundation elements and slope displacements shall be determined using empirical or elastoplastic analytical or numerical methods in the recognized literature by applying tsunami loads determined in Section 6.12.2 together with other applicable geotechnical and foundation loads required by this standard.

6.12.3 Alternative Foundation Performance-Based Design Criteria. In situ soil stresses from tsunami loads and effects shall be included in the calculation of foundation pressures. For local coseismic tsunami hazards that occur as a result of a local earthquake, the in situ soil and site surface condition at the onset of tsunami loads shall be those existing at the end of seismic shaking, including liquefaction, lateral spread, and fault rupture effects.

Building foundations shall provide sufficient capacity and stability to resist structural loads and the effects of general erosion and scour in accordance with the recognized literature. For Tsunami Risk Category IV buildings and structures, it shall be permitted to evaluate the overall performance of the foundation system for potential pore pressure softening by performing a two- or three-dimensional tsunami-soil-structure interaction numerical modeling analysis. The results shall be evaluated to demonstrate consistency with the structural performance acceptance criteria in Section 6.8. For Tsunami Risk Category IV buildings and structures, an independent peer review shall be conducted as part of a review of the performance-based design by the Authority Having Jurisdiction.

6.12.4 Foundation Countermeasures. Fill, protective slab on grade, geotextiles and reinforced earth systems, facing systems, and ground improvement shall be permitted to reduce the effects of tsunamis.

6.12.4.1 Fill. Fill used for structural support and protection shall be placed in accordance with ASCE 24 (2005), Sections 1.5.4 and 2.4.1. Structural fill shall be designed to be stable during inundation and to resist the loads and effects specified in Section 6.12.2.

6.12.4.2 Protective Slab on Grade. Exterior slabs on grade shall be assumed to be uplifted and displaced during the Maximum Considered Tsunami unless determined otherwise by site-specific design analysis based upon recognized literature.

Protective slabs on grade used as a countermeasure shall at a minimum have the strength necessary to resist the following loads:

1. Shear forces from sustained flow at maximum tsunami flow velocity, $u_{\max}$, over the slab on grade;
2. Uplift pressures from flow acceleration at upstream and downstream slab edges for both inflow and return flow;
3. Seepage flow gradients under the slab if the potential exists for soil saturation during successive tsunami waves;
4. Pressure fluctuations over slab sections and at joints;
5. Pore pressure increases from liquefaction and from the passage of several tsunami waves; and
6. Erosion of substrate at upstream, downstream, and flow parallel slab edges, as well as between slab sections.

6.12.4.3 Geotextiles and Reinforced Earth Systems. Geotextiles shall be designed and installed in accordance with manufacturers' installation requirements and as recommended in the recognized literature. Resistance factors required in Section 6.12.1 shall be provided for bearing capacity, uplift, lateral pressure, internal stability, and slope stability.

The following reinforced earth systems shall be permitted to be used:

1. Geotextile tubes constructed of high-strength fabrics capable of achieving full tensile strength without constricting deformations when subject to the design tsunami loads and effects;
2. Geogrid earth and slope reinforcement systems that include adequate protection against general erosion and scour, and a maximum lift thickness of 1 ft (0.3 m) and facing protection; and
3. Geocell earth and slope reinforcement erosion protection system designs, including an analysis to determine anticipated performance against general erosion and scour if no facing is used.

6.12.4.4 Facing Systems. Facing systems and their anchorage shall be sufficiently strong to resist uplift and displacement during design load inundation. The following facing methods for reinforced earth systems shall be permitted to be used:

1. Vegetative facing for general erosion and scour resistance where tsunami flow velocities are less than 12.5 ft/s (3.81 m/s). Design shall be in accordance with methods and requirements in the recognized literature.
2. Geotextile filter layers, including primary filter protection of countermeasures using a composite grid assuming high contact stresses and high-energy wave action design criteria in AASHTO M288-06, including soil retention, permeability, clogging resistance, and survivability.
3. Mattresses providing adequate flexibility and including energy dissipation characteristics. Edges shall be embedded to maintain edge stability under design inundation flows.
4. Concrete facing provided in accordance with protective slab on grade countermeasures in Section 6.12.4.2 and containing adequate anchorage to the reinforced earth system under design inundation flows.
5. Stone armoring and riprap provided to withstand tsunami shall be designed as follows: Stone diameter shall not be less than the size determined according to design criteria based on tsunami inundation depth and currents using design criteria in the recognized literature. Where the maximum Froude number, F_r , is 0.5 or greater, the high-velocity

turbulent flows associated with tsunamis shall be specifically considered, using methods in the recognized literature.

Subject to independent review, it shall be permitted to base designs on physical or numerical modeling.

6.12.4.5 Ground Improvement. Ground improvement countermeasures shall be designed using soil–cement mixing to provide nonerodible scour protection per Section 6.12.2.4 and at minimum provide soil–cement mass strength reinforcement of 100 psi (0.69 MPa) average unconfined compressive strength.

6.13 STRUCTURAL COUNTERMEASURES FOR TSUNAMI LOADING

The following countermeasures shall be permitted to reduce the structural effects of tsunamis.

6.13.1 Open Structures. Open Structures shall not be subject to Load Case 1 of Section 6.8.3.1. The load effect of debris accumulation against or within the Open Structure shall be evaluated by assuming a minimum closure ratio of 50% of the inundated projected area along the perimeter of the Open Structure.

6.13.2 Tsunami Barriers. Tsunami barriers used as an external perimeter structural countermeasure shall be designed consistent with the protected structure performance objectives to jointly achieve the performance criteria. These criteria include barrier strength, stability, slope erosion protection, toe scour, and geotechnical stability requirements and barrier height and footprint to fully prevent inundation during the Maximum Considered Tsunami. Where a barrier is designed to be overtopped by the design event or intended to provide only partial impedance of the design event, the protected structure and its foundation shall be designed for the residual inundation resulting from the design event. The foundation system treatment requirements in Section 6.12 of this chapter shall also be applied.

6.13.2.1 Information on Existing Buildings and Other Structures to Be Protected. As-built information on building configuration, building components, site, and foundation shall be permitted to be evaluated in accordance with ASCE 41 (2014), Chapters 9 through 11.

6.13.2.2 Site Layout. The spatial limits of the layout of tsunami barriers shall include the following:

1. The tsunami barrier shall be set back from the protected structure for perimeter protection. Any alignment change

shall have a minimum radius of curvature equal to at least half the maximum inundation depth.

2. For overtopping or partial impedance to inundation, at a minimum the barrier limits shall protect the structure from inundation flow based on an approach angle of ± 22.5 degrees from the shoreline. The flow approach angle shall be evaluated in accordance with Sections 6.8.6.1 and 6.8.6.2.

6.14 TSUNAMI VERTICAL EVACUATION REFUGE STRUCTURES

Tsunami Vertical Evacuation Refuge Structures designated as a means of alternative evacuation by the Authority Having Jurisdiction shall be designed in accordance with the additional requirements of this section.

6.14.1 Minimum Inundation Elevation and Depth. Tsunami refuge floors shall be located not less than the greater of 10 ft (3.05 m) or one-story height above 1.3 times the Maximum Considered Tsunami inundation elevation at the site as determined by a site-specific inundation analysis, as indicated in Fig. 6.14-1. This same Maximum Considered Tsunami site-specific inundation elevation, factored by 1.3, shall also be used for design of the Tsunami Vertical Evacuation Refuge Structure in accordance with Sections 6.8 to 6.12.

6.14.2 Refuge Live Load. An assembly live load, L_{refuge} , of 100 psf (4.8 kPa) shall be used in any designated evacuation floor area within a tsunami refuge floor level.

6.14.3 Laydown Impacts. Where the maximum inundation depth exceeds 6 ft (1.83 m), the laydown impact of adjacent pole structures collapsing onto occupied portions of the building shall be considered.

6.14.4 Information on Construction Documents. Construction documents shall include tsunami design criteria and the occupancy capacity of the tsunami refuge area. Floor plans shall indicate all refuge areas of the facility and exiting routes from each area. The latitude and longitude coordinates of the building shall be recorded on the construction documents.

6.14.5 Peer Review. Design shall be subject to independent peer review by an appropriately licensed design professional who shall present a written report to the Authority Having Jurisdiction as to the design's compliance with the requirements of this standard.

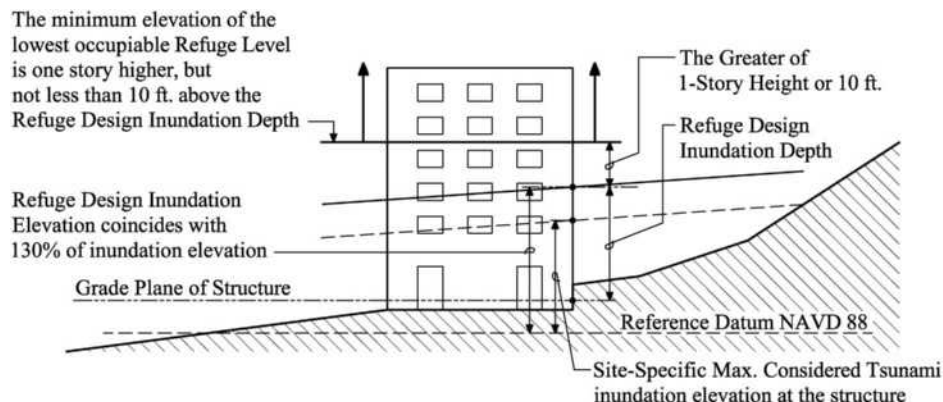


FIGURE 6.14-1 Minimum Refuge Level Elevation [1 ft=0.305 m]

6.15 DESIGNATED NONSTRUCTURAL COMPONENTS AND SYSTEMS

6.15.1 Performance Requirements. Designated nonstructural components and systems in structures located in the Tsunami Design Zone shall be either protected from tsunami inundation effects or positioned in the structure above the inundation elevation of the Maximum Considered Tsunami, such that the designated nonstructural components and systems will be capable of performing their critical function during and after the Maximum Considered Tsunami. Tsunami barriers used as inundation protection shall have a top-of-wall elevation that is not less than 1.3 times the maximum inundation elevation at the barrier. The tsunami barrier shall also satisfy the requirements of Section 6.13. Alternatively, it shall be permitted to design the designated nonstructural components and systems directly for tsunami effects, provided that inundation would not inhibit them from performing their critical function during and after the Maximum Considered Tsunami.

6.16 NONBUILDING TSUNAMI RISK CATEGORY III AND IV STRUCTURES

6.16.1 Requirements for Tsunami Risk Category III Nonbuilding Structures. Tsunami Risk Category III nonbuilding structures located in the Tsunami Design Zone shall be either protected from tsunami inundation effects or designed to withstand the effects of tsunami loads in accordance with Section 6.8 of this chapter and in accordance with the specific performance requirements of Section 6.8.3. Tsunami barriers used as inundation protection shall have a top-of-wall elevation that is not less than 1.3 times the maximum inundation elevation at the barrier. The tsunami barrier shall also satisfy the requirements of Section 6.13.

6.16.2 Requirements for Tsunami Risk Category IV Nonbuilding Structures. Tsunami Risk Category IV designated nonstructural systems in nonbuilding structures located in the Tsunami Design Zone shall be (1) protected from tsunami inundation effects, (2) positioned above 1.3 times the inundation elevation of the Maximum Considered Tsunami in such

a manner that the Tsunami Risk Category IV nonbuilding structure will be capable of performing its critical function during and after the Maximum Considered Tsunami, or (3) designed to withstand the effects of tsunami loads in accordance with Section 6.8 of this chapter and the specific performance requirements of Section 6.8.3. Tsunami barriers used as inundation protection shall have a top-of-wall elevation that is not less than 1.3 times the maximum inundation elevation at the barrier. The tsunami barrier shall also satisfy the requirements of Section 6.13.

6.17 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

This section lists the consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter. Those referenced documents identified by an asterisk (*) are not consensus standards; rather, they are documents developed within the industry and represent acceptable procedures for design and construction to the extent referred to in the specified section.

AASHTO Guide Specifications and Commentary for Vessel Collision Design of Highway Bridges, 2nd Ed., American Association of State Highway and Transportation Officials, 2009, with 2010 Interim Revisions.

Cited in: Section 6.11.5

AASHTO M288-06, *Standard Specification for Geotextile Specification for Highway Applications*, American Association of State Highway and Transportation Officials, 2006.

Cited in: Section 6.12.4.4

ASCE/SEI 24-05, *Flood Resistant Design and Construction*, American Society of Civil Engineers, 2005.

Cited in: Section 6.12.4.1

ASCE/SEI 41-13, *Seismic Evaluation and Retrofit of Existing Buildings*, American Society of Civil Engineers, 2014.

Cited in: Sections 6.8.3.5.2.2 and 6.13.2.1

***NOAA Technical Memorandum OAR PMEL-135**, *Standards, Criteria, and Procedures for NOAA Evaluation of Tsunami Numerical Models*, National Oceanic and Atmospheric Administration, Pacific Marine Environmental Laboratory, 2007.

Cited in: Section 6.7.6.7.2